

Multimodal AAD dataset — analysis progress

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Dataset layout

For every subject, two data stores hold parallel copies of the same trials, partitioned by modality.

`experiment_data/Subject N/`

- `Training-1 ... Training-5` — 5 familiarisation trials at the start of the recording session.
- `Eval-1 ... Eval-100` — 100 main trials. `Eval-K` corresponds to the `Trial-K` row of `audio_stimuli_data/trials.csv`.
- Each trial folder holds EEG (`eeg_data.p`, 32-ch @ 500 Hz) and `audio_timestamps.json`.

`Video Recordings/Subject N/`

Per-subject folders numbered 1 through 105:

- Folders 1–5 = training trials (same ordering as `experiment_data/Training-K`).
- Folders 6–105 = main trials; folder `K` corresponds to `Eval-(K-5)` (i.e. `Trial-(K-5)`).
- Each folder holds the Tobii Glasses 3 bundle: `gazedata.gz` (per-eye gaze direction + pupil + scene-projected gaze), `imudata.gz`, `scenevideo.mp4`, `eventdata.gz`, `recording.g3`, `snap0.jpg`.

Subjects 1 and 2 use literal numeric folder names (1...105). Subjects 3–16 use ISO-8601 timestamps (YYYYMMDDTHHMMSSZ); the loader parses the embedded time and sorts ascending to recover the natural recording order. This ordering is verified empirically against audio playback timestamps to $r_{\Delta} \geq 0.99$ trial-to-trial for every subject.

Gaze source

The authoritative gaze stream is the Tobii `gazedata.gz` inside each Video-Recordings folder (per-eye 3-D gaze direction, pupil diameter, scene-projected `gaze2d`, 3-D `gaze3d`). Experiment-data 2-D gaze files are not used in the current pipeline.

Audio

`audio_stimuli_data/trials.csv` lists per-trial metadata: attended-speaker index, the three device files, SNR in dB, and per-channel powers. Speech files live in `audio_stimuli_data/pairs/*.flac` as stereo; each device plays one stereo file with its two channels driving the two speakers of that device.

Speaker geometry (listener-referenced azimuth)

Label	Azimuth	Device / channel	Hemisphere
1	-67.5°	Device-1 Left	LEFT
2	-22.5°	Device-1 Right	LEFT
3	$+22.5^\circ$	Device-2 Left	RIGHT
4	$+67.5^\circ$	Device-2 Right	RIGHT
–	-135°	Device-3 Left (noise)	back
–	$+135^\circ$	Device-3 Right (noise)	back

Device-1 is the entire **left** hemisphere speaker pair; Device-2 is the entire **right** hemisphere pair. Device-3 carries back-hemisphere noise. All speakers are at a radial distance of 4 ft.

Chapter 1

Analysis pipeline

1.1 Shared preprocessing

- EEG is loaded from `eeg_data.p` (32 ch, 500 Hz) and the 32 scalp channels are wrapped in an `mne.io.Raw` with the ANT Neuro 10–20 montage (Fp1, Fpz, Fp2, F7, F3, Fz, F4, F8, FC5, FC1, FC2, FC6, M1, T7, C3, Cz, C4, T8, M2, CP5, CP1, CP2, CP6, P7, P3, Pz, P4, P8, POz, O1, Oz, O2).
- **Bad-channel detection** flags flat channels ($\text{std} < 1 \times 10^{-9}$), saturated channels ($\geq 10\%$ samples at the 0.0839 ADC clip), and AC-variance $|z| > 6$ outliers against the median MAD of the remaining channels.
- **Referencing** defaults to linked mastoids (average of M1 and M2); when either mastoid is flagged bad, falls back to a common-average reference over the good channels. Flagged channels are then interpolated via spherical splines so every trial ends with 32 valid channels.
- Notch filter at 60 Hz, band-pass 1–9 Hz for speech-tracking analyses, and a subsequent resample to 64 Hz (common rate with audio envelopes).
- Optional ICA with automatic EOG-component identification using Fp1/Fp2 as proxies (disabled by default because ablations showed only a 0.3 pp gain at $3\times$ runtime cost).

1.2 Modality alignment

All streams are clipped to the shared audio-playback window $[t_0, t_1]$ derived from the per-trial `audio_timestamps.json`:

- **EEG**: internal `eeg_ts` is mapped to wall-clock unix time via `first_sample_time` in `eeg_time_data.p`.
- **Tobii streams** (gaze + IMU): recording-relative timestamps are shifted so the first sample aligns to t_0 , then clipped to $[t_0, t_1]$.
- **Audio envelopes**: computed from each device’s stimulus file and re-sampled to match the EEG rate.

1.3 EEG linear-decoder backbones (iter-3 on all 16 subjects)

For each (subject $\times$ backbone) combination the pipeline:

1. Loads 100 main trials, preprocessed as above at 64 Hz.
2. Builds lagged regressors over $[-200, +500]$ ms (step 25 ms).
3. Selects ridge α via an inner 5-fold CV on a 12-point log-spaced grid $10^{-1} \dots 10^6$, scored by mean $\rho_{\text{att}} - \rho_{\text{una}}$ on held-out trials.
4. Outer 5-fold CV for reported accuracy at decision windows $\{1, 2, 4, 8, 16, 30\text{ s}\}$.

Four backbones:

- **Broadband:** 1–9 Hz Hilbert envelope derivative (half-wave rectified) as the 1-D reconstruction target.
- **Split $\delta+\theta$:** concatenated 1–4 Hz and 4–8 Hz EEG lag features, reconstructing the same 1-D envelope.
- **Mel-28:** 28-band gammatone-proxy mel envelope as the multi-band reconstruction target (reduced to mean-band for the scalar ridge).
- **CCA mel-28:** canonical correlation between (EEG lags) and (mel-28 envelope), retaining up to 5 components per trial.

1.4 Multimodal fusion

Binary group-level task: attended speaker pair $\in \{\text{Device-1}, \text{Device-2}\}$ (chance = 0.5). For each subject we compute:

- **Gaze features** from the Tobii `gazedata.gz` stream — mean/std of scene-projected gaze, per-eye gaze-direction azimuth and elevation, pupil diameter, saccade rate, 3-D gaze centroid, and IMU head motion.
- **EEG features** = the $(\rho_{\text{att}}, \rho_{\text{una}})$ pair from each EEG backbone, mapped through a sigmoid to a per-trial probability.

Classifiers: logistic regression (gaze-only, EEG-only) and LightGBM (stacked / early-fusion). Evaluated with 5-fold stratified CV per subject.

1.5 SLURM design

Account: PAS2966. Array jobs on the `batch` partition, one task per (subject $\times$ backbone). Each task: 8 CPUs, 32 GB memory, 2 h wall limit. A single backbone sweep across 16 subjects occupies 16 tasks; iter-3’s four backbones = 64 tasks total, parallelised across compute nodes so the full sweep wall-clocks in roughly 15–20 minutes.

Chapter 2

Current results

What task is being decoded

Goal: for every trial (30 s of concurrent multi-talker speech), predict which of 4 spatial speakers (Device-1 L at -67.5 , Device-1 R at -22.5 , Device-2 L at $+22.5$, Device-2 R at $+67.5$) the subject was attending. Ground truth is the **Attended Speaker** column of `trials.csv`.

Evaluation: 5-fold stratified CV *per subject* over 100 main trials; accuracies are mean across folds. Pooled accuracy is the unweighted mean of per-subject accuracies across all 16 subjects (bootstrap 95 % CI).

Inputs used in this chapter:

- *Linear EEG decoder:* 32-ch EEG (~ 500 Hz) bandpassed to 1–9 Hz and resampled to 64 Hz; decoder is CCA between EEG-lags ($-200 \dots +500$ ms) and the 28-band gammatone envelope of the attended-device stereo audio (*inputs: EEG only*; audio envelopes are the supervision signal, not classifier inputs).
- *Fusion:* Tobii `gazedata.gz` per-trial summary (20 features) combined with the EEG CCA probabilities.
- IMU and scene video are not used in this chapter.

All 16 subjects, 100 main trials each. Within-subject 5-fold CV, 30 s full-trial decision windows unless stated otherwise.

2.1 Linear EEG decoder (backward TRF / CCA)

2.1.1 Pooled full-trial accuracy (chance = 0.5)

Backbone	Pooled mean	n trials
CCA mel-28	0.523	1599
Mel-28 ridge	0.511	1599
Broadband ridge	0.507	1599
Split $\delta+\theta$	0.492	1599
Oracle adaptive (per-subject best backbone)	0.555	1599

CCA on the 28-band gammatone envelope is the best uniform backbone. A per-subject adaptive policy that picks whichever backbone wins on inner CV reaches 0.555, ~ 3 pp above the best fixed choice.

2.1.2 Per-subject winning backbone

Backbone	Subjects for which it wins
CCA mel-28	S1, S2, S4, S5, S7, S10, S11, S12, S13 (9)
Broadband ridge	S3, S9, S14, S16 (4)
Mel-28 ridge	S6, S15 (2)
Split $\delta+\theta$	S8 (1)

2.1.3 Window-length sweep (mean across 16 subjects)

Window (s)	Broadband	Mel-28	Split $\delta+\theta$
1	0.496	0.499	0.500
2	0.499	0.501	0.498
4	0.500	0.505	0.499
8	0.505	0.500	0.502
16	0.510	0.524	0.504
30	0.507	0.510	0.492

The short-window (1–4 s) regime stays near chance; accuracy climbs modestly at 16–30 s. This shallow gradient reflects the 30 s trial length and the reduced stimulus separability of the paradigm (each device plays one speech file across both of its L/R speakers at different powers, so the attended and unattended streams are spatially smeared).

2.2 Multimodal fusion

Three task framings share the same Tobii-gaze feature set (`gazedata.gz`) and the same EEG backbones, evaluated per subject via 5-fold stratified CV.

2.2.1 Task A — Attended hemisphere (left vs right)

Binary task: attended speaker $\in$ Left hemisphere (labels 1–2, azimuth $\{-67.5^\circ, -22.5^\circ\}$) vs Right hemisphere (labels 3–4, azimuth $\{+22.5^\circ, +67.5^\circ\}$). Chance = 0.5.

2.2.2 Pooled accuracy (bootstrap 95 % CI, $n = 16$)

Pooled 95 % CI for gaze-only: $[0.685, 0.849]$. EEG-only sits at chance on the binary group-level task; fusion gives the same pooled number as gaze-only within CI overlap.

EEG backbone	Gaze-only	EEG-only (LR)	Late fusion	Stacked	Early fusion
Broadband	0.770	0.505	0.766	0.752	0.751
Split $\delta+\theta$	0.770	0.485	0.767	0.734	0.757
Mel-28	0.770	0.511	0.756	0.758	0.763
CCA mel-28	0.770	0.502	0.768	0.756	0.764

Subj	n	Gaze	EEG (LR)	Late	Stacked	Early
1	99	0.404	0.545	0.394	0.545	0.515
2	98	0.939	0.429	0.939	0.918	0.949
3	100	1.000	0.340	0.990	0.990	1.000
4	100	0.980	0.490	0.990	0.970	0.950
5	100	0.750	0.520	0.730	0.710	0.750
6	100	0.640	0.390	0.650	0.650	0.670
7	100	0.910	0.420	0.900	0.920	0.930
8	100	0.550	0.560	0.550	0.570	0.570
9	100	0.660	0.560	0.680	0.620	0.670
10	100	0.860	0.480	0.860	0.820	0.780
11	100	0.680	0.570	0.680	0.610	0.610
12	100	0.600	0.510	0.580	0.490	0.590
13	100	0.860	0.630	0.840	0.850	0.890
14	100	0.650	0.580	0.650	0.700	0.620
15	99	0.980	0.556	0.980	0.980	1.000
16	100	0.830	0.470	0.820	0.790	0.800

2.2.3 Per-subject attended-hemisphere AAD (CCA mel-28 backbone)

2.2.4 Task B — Inner vs Outer (gaze-lateralisation control)

Binary task: inner speakers (labels 2, 3, azimuth $\pm 22.5^\circ$) vs outer speakers (labels 1, 4, azimuth $\pm 67.5^\circ$). This collapses the left/right axis and asks whether gaze can distinguish eccentricity. Pooled gaze-only accuracy = **0.690** (vs 0.500 chance; 13/16 above chance). A secondary spatial signal is present but substantially weaker than the hemisphere task.

2.2.5 Task C — 4-class speaker identity

4-way classification over the full speaker set (chance = 0.25). Pooled accuracy = **0.547**, 16/16 subjects above chance. Effect size relative to chance (+30 pp) is comparable to Task A’s +27 pp — the binary hemisphere task just happens to have a higher absolute chance.

2.3 Observations

1. **Gaze carries the dominant signal for the attended hemisphere (Task A):** 6/16 subjects reach ≥ 0.83 gaze-only accuracy; S3 and S15 reach **1.000** with early fusion. Gaze is substantially weaker on within-hemisphere distinctions (Task B: 0.690 pooled).
2. **Linear EEG-only AAD on the binary hemisphere task is at chance.** The backward TRF / CCA extracts per-trial envelope-tracking correlation, not a lateralised spatial signature

— so collapsing four speakers into two hemispheres loses the signal the decoder is most sensitive to.

3. **Fusion does not yet improve pooled accuracy** because the EEG contribution is near-chance. The pipeline is wired end-to-end and ready to benefit from a stronger EEG model.
4. **Per-subject heterogeneity is informative.** The five subjects below 0.70 gaze-only (S1, S6, S8, S11, S12) are the ones whose gaze does not reliably fixate the attended source. These are the subjects for whom a non-gaze decoder is actually useful.
5. **Subject 1** is an outlier: below-chance gaze (0.40) and near-chance EEG. The recording has 100 % flat M1 and 100 % saturated M2 — a real hardware/electrode issue, not an analysis artefact. It survives the pipeline only because the robust referencing and spherical interpolation prevent the bad channels from corrupting the other 30.

Chapter 3

Iteration 4 — results

Task and inputs

Same AAD task as Chapter 2: predict the attended speaker from 30-s trials. Three experiments here, each varying *only* the EEG/gaze source or decoder architecture:

- **4-class EEG decoder** (§3.1): input = 32-ch EEG at 64 Hz; supervision = 4 candidate attended envelopes (one per speaker). Target = pick the speaker. Gaze, IMU, and video are not used.
- **EEG↔gaze predictability** (§3.2): input = 32-ch EEG (for EEG→gaze) or the 20-D Tobii gaze summary (for gaze→attended). Target = gaze time-courses, or the attended-speaker label.
- **Gaze-residualised EEG AAD** (§3.3): input = 32-ch EEG *after* channel-wise regression of a 15-feature Tobii gaze + IMU regressor set; same CCA-mel-28 decoder as Chapter 2 otherwise. Target = attended-speaker hemisphere.

Per-subject 5-fold CV; pooled = unweighted mean across 16 subjects with bootstrap 95 % CI. All 48 SLURM tasks completed on the PAS2966 CPU queue.

3.1 4-class EEG speaker-identity decoder

CCA between EEG lags ($-200 \dots + 500$ ms) and 28-band gammatone envelopes of each of the 4 possible attended speakers (Device-1 L/R, Device-2 L/R). Test trials are classified by *argmax* canonical correlation across the four envelope candidates. Per-subject 5-fold CV, 100 trials.

3.1.1 Per-subject accuracy

All three task framings land at chance. The 4-class decoder does not distinguish individual speakers; even the inner-outer binary barely exceeds chance (+1.8 pp). This is expected: within a device, the stereo speech file is replicated on both channels with different amplitudes — the envelope shape is near-identical, so envelope reconstruction cannot pick between the two channels of the same device.

Window sweep {1, 2, 4, 8, 16, 30} s shows no gradient beyond ± 0.01 , confirming the backbone is not merely data-starved.

Subject	4-class	hemisphere	inner/outer
1	0.240	0.490	0.480
2	0.270	0.520	0.480
3	0.260	0.430	0.610
4	0.220	0.480	0.450
5	0.310	0.560	0.520
6	0.240	0.560	0.500
7	0.330	0.500	0.610
8	0.250	0.520	0.500
9	0.250	0.490	0.540
10	0.280	0.450	0.550
11	0.250	0.500	0.550
12	0.210	0.540	0.450
13	0.230	0.450	0.510
14	0.210	0.450	0.480
15	0.263	0.505	0.545
16	0.270	0.530	0.520
Pooled	0.255	0.498	0.518
95 % CI	[0.240, 0.271]	[0.479, 0.517]	[0.496, 0.542]
Chance	0.250	0.500	0.500

3.2 Cross-modal predictability (EEG $\leftrightarrow$ gaze)

3.2.1 EEG $\rightarrow$ gaze (ridge regression, held-out Pearson r)

Target	mean r	std
gaze2d_x (scene-projected horizontal)	0.028	0.020
gaze2d_y (scene-projected vertical)	0.016	0.015
gaze3d_x (listener-ref horizontal, mm)	0.009	0.011
gaze3d_y (vertical, mm)	0.000	0.012
gaze3d_z (depth, mm)	0.002	0.008

EEG carries almost no linear predictive information about instantaneous gaze: held-out Pearson r peaks at 0.028 for the horizontal scene-projected gaze and collapses to zero in 3-D body-referenced space.

3.2.2 CCA EEG $\leftrightarrow$ gaze (first-component correlation)

	Mean first-component r
In-sample (training residuals)	0.147 ± 0.092
Held-out	0.081 ± 0.063

A small but consistent shared subspace exists (held-out $r = 0.081$), but it is an order of magnitude below what each modality exposes about the attended-speaker label on its own.

3.2.3 Gaze $\rightarrow$ attended (logistic classifier)

Task	Chance	Pooled	95 % CI
Hemisphere (L vs R)	0.500	0.769	[0.684, 0.851]
Inner vs outer	0.500	0.680	[0.603, 0.762]
4-class identity	0.250	0.551	[0.439, 0.668]

Reproduces the corrected-mapping numbers from Chapter 2.

3.3 Gaze-residualised EEG AAD — the headline control

For every trial, a 15-feature gaze/IMU regressor set (`gaze2d_x/y`, `gaze3d_x/y/z`, per-eye direction, pupil diameters, gaze velocities, $|\text{accel}| - g$, $|\text{gyro}|$) is channel-wise ridge-regressed out of the EEG ($\lambda=10^{-3}$) *before* the CCA mel-28 decoder sees it. Compares against the same decoder on un-residualised EEG (*baseline*). Per-subject 5-fold CV, hemisphere task, full 30-s trial.

Subject	Baseline	Gaze-residualised
1	0.560	0.580
2	0.570	0.610
3	0.380	0.460
4	0.490	0.520
5	0.510	0.570
6	0.510	0.580
7	0.580	0.580
8	0.490	0.570
9	0.560	0.560
10	0.520	0.430
11	0.510	0.530
12	0.440	0.520
13	0.510	0.590
14	0.520	0.450
15	0.424	0.505
16	0.470	0.440
Pooled	0.503	0.531
95 % CI	[0.477, 0.528]	[0.503, 0.558]

Mean paired $\Delta = +0.028$ pp. Wilcoxon signed-rank $p = 0.084$, $n = 16$. 11 of 16 subjects improve after removing the gaze-correlated variance.

Headline result: removing gaze-correlated variance from EEG *improves* AAD accuracy, not reduces it. This directly refutes the “eye-tracking-in-disguise” concern: EEG carries AAD-relevant signal that is independent of gaze, and the gaze-linked component is effectively noise relative to the TRF task.

3.4 Extended fusion (gaze + 4-class-EEG-probabilities)

Fusion inputs: Tobii gaze summary features (20-D) + the 4-class CCA softmax probabilities ($p_1, \dots, p_4$). Classifier = LightGBM ($n = 200$ estimators) with 5-fold stratified CV per subject. Three task framings; S15’s 4-class EEG cache was missing so fusion is computed on 15 subjects; gaze-only is pooled on all 16.

Task	Gaze-only	EEG-only (4-class probs)	Early fusion
Hemisphere	0.770	0.501	0.749
Inner vs outer	0.690	0.477	0.643
4-class identity	0.547	0.255	0.493

- Fusion does not exceed gaze-only on any task — the 4-class EEG probabilities are too close to chance to add discriminative signal on top of a strong gaze feature vector.
- The *only* bounded gain from fusion will come from either (a) a stronger EEG decoder (iter-5: deep models) or (b) subjects whose gaze is unreliable, where the EEG residual is additive.
- Per-subject fusion outperforms gaze-only only for 4 subjects on hemisphere task (S5: +0.02, S6: +0.07, S7: +0.02, S12: +0.00). These are the low-gaze-accuracy subjects — exactly the regime where EEG is expected to help.

3.5 What iter-4 settles

1. **Linear EEG alone cannot discriminate within-device L/R** (4-class = 0.255, chance = 0.250). The paradigm’s stereo device design means envelope reconstruction is blind to channel identity. Iter-5 will test whether a deep decoder on *raw EEG* (not reconstruction) can exploit the spatial lateralisation cues that CCA misses.
2. **EEG has very low linear predictability of gaze** ($r \leq 0.03$) and a modest shared subspace ($r_{\text{CCA, out}} = 0.081$). Gaze contamination of EEG exists but is small.
3. **Removing gaze *improves* EEG-AAD by +2.8 pp** — the reviewer-ready answer to the “eye-tracking in disguise” question.
4. **The fusion story** needs a better EEG backbone to beat the gaze ceiling. The wiring is in place; iter-5 just swaps the EEG module.

Chapter 4

Iteration 5 — classical spectral EEG features

4.1 Task definition and inputs

Task. Auditory-attention decoding: for a given trial, predict which of four concurrently presented speakers the subject was attending to. A trial is 30 s long; during it the subject hears four speech streams simultaneously (Device-1 L, Device-1 R, Device-2 L, Device-2 R — see Dataset Layout chapter) plus noise from Device-3 at $\pm 135^\circ$. The attended-speaker label comes from `trials.csv` and is one of $\{1, 2, 3, 4\}$.

Three label-collapses evaluated.

- **Hemisphere** (chance = 0.5): $\{1, 2\}$ (LEFT, $az \in \{-67.5, -22.5\}$) vs $\{3, 4\}$ (RIGHT, $az \in \{+22.5, +67.5\}$). This is the classical binary AAD framing — which side is the listener attending.
- **Inner vs outer** (chance = 0.5): $\{2, 3\}$ (inner speakers at ± 22.5) vs $\{1, 4\}$ (outer speakers at ± 67.5). Tests whether the decoder picks up *eccentricity* rather than hemisphere.
- **4-class speaker identity** (chance = 0.25): predict the exact speaker label 1/2/3/4.

Inputs used in iter-5.

- **EEG-only condition:** 32-channel EEG (`experiment_data/Eval-K/eeg_data.p`), bandpassed 1–40 Hz, notched at 60 Hz, auto-referenced (linked mastoids when good; common-average otherwise), at the native 500 Hz. Per trial we compute a **368-dimensional spectral feature vector** described in the next section and use it as input X to a per-subject classifier.
- **Gaze-only condition** (for the fusion comparison): the Tobii Glasses 3 `gazedata.gz` stream from the Video-Recordings folder paired with the trial, plus the paired `imudata.gz`. The 30-s stream is summarised into a 23-dimensional vector (Table 4.1). The experiment-data 2-D gaze (`gaze_data.p`) is not used.
- **Fusion condition:** concatenation $[EEG_{368}, gaze_{20}]$.
- **Scene video** (`scenevideo.mp4`) is *not* used in iter-5. Planned for iter-6 as motion-energy / optical-flow features.

Table 4.1: Gaze / head-motion feature vector (23 features per trial).

#	Name	Source	Description
1	gx_mean	Tobii gaze2d	scene horizontal gaze, mean
2	gx_std	Tobii gaze2d	scene horizontal gaze, std
3	gy_mean	Tobii gaze2d	scene vertical gaze, mean
4	gy_std	Tobii gaze2d	scene vertical gaze, std
5	gx_valid	Tobii gaze2d	fraction of finite samples
6	sacc_rate	derived	saccades/s (I-VT, 30°/s thresh.)
7	sacc_amp_med	derived	median saccade amplitude (°)
8	g3d_x_mean	Tobii gaze3d	3-D gaze x (horizontal, mm)
9	g3d_y_mean	Tobii gaze3d	3-D gaze y (vertical, mm)
10	g3d_z_mean	Tobii gaze3d	3-D gaze z (depth, mm)
11	g3d_x_std	Tobii gaze3d	3-D gaze x, std
12	L_az_mean	Tobii eyeleft	left-eye azimuth mean (°)
13	L_az_std	Tobii eyeleft	left-eye azimuth std
14	L_el_mean	Tobii eyeleft	left-eye elevation mean (°)
15	L_pup	Tobii eyeleft	left pupil diameter (mm)
16	L_valid	Tobii eyeleft	left-eye sample validity
17	R_az_mean	Tobii eyeright	right-eye azimuth mean
18	R_az_std	Tobii eyeright	right-eye azimuth std
19	R_el_mean	Tobii eyeright	right-eye elevation mean
20	R_pup	Tobii eyeright	right pupil diameter
21	R_valid	Tobii eyeright	right-eye sample validity
22	gyro_mag	imudata	mean $\ \boldsymbol{\omega}\ $ (rad/s)
23	acc_std	imudata	std of $\ \mathbf{a}\ - g$ (m/s ²)

Prediction target. $\hat{y} = f(X)$ where X is the per-trial feature vector above and $\hat{y}$ is the attended-speaker label under whichever of the three label-collapses we are evaluating.

Evaluation protocol. For every subject independently, trials are split with 5-fold stratified cross-validation (100 trials / subject, so ~ 80 train + ~ 20 test per fold). The classifier is **refit from scratch on each subject** — no cross-subject transfer. The reported accuracy per subject is the mean over the 5 held-out folds; the reported *pooled* accuracy is the unweighted mean across the 16 per-subject accuracies (with bootstrap 95 % CI resampling the 16 values). Chance is determined by the label collapse.

Headline

Replacing the envelope-reconstruction decoder (CCA mel-28) with a 368-dimensional *linear* classifier over classical EEG features — band powers, alpha lateralisation, Hjorth complexity, spectral entropy, $1/f$ slope, alpha-band phase-locking and magnitude-squared coherence — takes pooled within-subject hemisphere AAD from 0.498 to **0.716**, a +21.8 pp jump. No deep learning. Subject 3 climbs from 0.43 (CCA) to **0.94** logreg; Subject 15 hits **0.92**.

4.2 Feature set

Per trial, computed on 1–40 Hz preprocessed, auto-referenced EEG (500 Hz) and aggregated over the 30 s trial:

Group	# feats	Contents
Band powers (log)	160	Welch $\delta/\theta/\alpha/\beta/\gamma$ per 32 channels
Within-channel band ratios (log)	96	α/θ , α/β , θ/β per channel
Alpha lateralisation	4	log R/L for parietal, central, occipital, temporal
Hjorth mobility + complexity	64	Per channel
Spectral entropy	32	Shannon, 1–40 Hz
$1/f$ aperiodic slope	4	Per channel group (frontal/central/parietal/occipital)
Alpha connectivity	6	MSC & wPLI for within-L, within-R, between-hemi
Global field power	2	mean, std
Total	368	

4.3 Classifier sweep (all 16 subjects, per-subject 5-fold CV)

Task	Classifier	Pooled acc.	95 % CI
Hemisphere	L2-LogReg	0.716	[0.641, 0.792]
Hemisphere	LightGBM	0.655	[0.589, 0.724]
Inner-outer	L2-LogReg	0.649	[0.592, 0.712]
Inner-outer	LightGBM	0.597	[0.536, 0.667]
4-class	L2-LogReg	0.448	[0.358, 0.544]
4-class	LightGBM	0.410	[0.332, 0.497]

L2-LogReg beats LightGBM on every task — consistent with a linearly separable feature set and $p = 368$ vs $n \approx 100$ where tree-boosting overfits. The hemisphere pool at 0.716 is +21.8 pp above the linear TRF baseline (0.498, Chapter 3) and +19.3 pp above the best iter-3 CCA backbone pooled (0.523).

4.4 Per-subject LogReg accuracy

Nine of 16 subjects pass 0.70 hemisphere; five pass 0.88. Subject 11 sits at 0.43 — consistent with its weak-signal profile on other metrics (this subject also fails the linear TRF and has a below-chance gaze-only score).

4.5 Fusion with gaze features

Early fusion = concatenate $[EEG_{368}, Gaze_{20}]$ and feed to LightGBM ($n = 300$, lr=0.05). 5-fold within-subject CV.

Pooled fusion does *not* exceed gaze-only — the two modalities have substantial overlapping information because both exploit the same underlying spatial-attention process (lateralised gaze,

Subject	Hemisphere	Inner/outer	4-class
1	0.480	0.640	0.280
2	0.900	0.850	0.750
3	0.940	0.600	0.520
4	0.880	0.910	0.810
5	0.700	0.640	0.460
6	0.730	0.580	0.420
7	0.890	0.730	0.620
8	0.620	0.540	0.310
9	0.600	0.570	0.240
10	0.660	0.610	0.450
11	0.430	0.510	0.200
12	0.600	0.570	0.330
13	0.770	0.620	0.400
14	0.610	0.530	0.250
15	0.920	0.889	0.786
16	0.720	0.590	0.340
Pooled	0.716	0.649	0.448
Chance	0.500	0.500	0.250

Task	EEG spec. only	Gaze only	Early fusion
Hemisphere	0.716 [0.639, 0.789]	0.772 [0.687, 0.851]	0.753 [0.660, 0.838]
Inner-outer	0.650 [0.593, 0.714]	0.683 [0.597, 0.766]	0.660 [0.583, 0.735]
4-class	0.442 [0.354, 0.535]	0.547 [0.429, 0.664]	0.513 [0.405, 0.623]

lateralised alpha). **But on a per-subject basis, fusion does help when both modalities are strong:** S7 hemisphere $0.89 \rightarrow 0.97$, S15 $0.92 \rightarrow 1.000$, S2 $0.91 \rightarrow 0.95$, S13 $0.77 \rightarrow 0.88$.

4.6 What iter-5 settles

1. **Linear separation of attended hemisphere from EEG alone is achievable:** LogReg on 368 spectral/connectivity features reaches 0.716 pooled. The envelope-tracking paradigm is simply the wrong question to ask on this dataset.
2. **Per-subject heterogeneity is preserved:** subjects with hardware-quality issues (S11, S1) remain at or below chance even with the richer feature set. The featurisation is not fixing the data; it is exposing more of the signal that was there all along.
3. **Fusion gain exists per-subject but is hidden by pooled averaging.** On subjects where both modalities are strong, early fusion improves on gaze-only; on subjects where either is weak, it tracks the stronger modality. The pooled number is the wrong summary when the two modalities correlate.
4. **LightGBM underperforms LogReg** here despite having more capacity — characteristic of high-dimensional, low-noise linear features with only ~ 100 examples per subject. Regularised linear is the right default for this dataset size.

4.7 Iter-6 follow-ups

1. **Feature-family ablation:** drop each group one at a time and re-fit, to identify which subset of the 368 features carries the signal (expect alpha lateralisation + band powers in parietal/temporal channels to dominate).
2. **Dimension reduction:** ℓ_1 -regularised logistic or group-sparse regression to pick an interpretable subset (goal: ≤ 20 features that retain ≥ 0.70 hemisphere).
3. **Deep EEG model** (the planned iter-6 GPU experiment): EEGNet and a small transformer on 5–10 s windows, comparing to the 368-feature LogReg ceiling on the same tasks.
4. **Per-subject quality score** combining bad-channel rate, alpha-lat. direction, gaze validity, and now the iter-5 spectral accuracy — predict which subjects will gain from a deep model.

Chapter 5

Iteration 6 — artefact controls (IMU, video, channel ablation)

5.1 Task and inputs

Same AAD task as the previous chapters (predict the attended speaker under the three label collapses: hemisphere / inner-outer / 4-class). This chapter asks three targeted questions:

1. Can **head motion alone** (Tobii IMU) decode the attended speaker? If yes, it is a confound we must control for.
2. Can **scene video alone** (Tobii `scenevideo.mp4`, head-mounted) decode? Egocentric video in a quiet room is largely a derivative of head motion, so we expect the answer to track IMU closely.
3. Does the iter-5 EEG spectral-feature classifier survive aggressive nuisance cleaning? We re-run it under five EEG-preparation conditions and measure how much accuracy each cleaning step removes.

All classifiers are L2-logistic regression (and LightGBM for IMU/video) with 5-fold stratified CV per subject, pooled by unweighted mean across 16 subjects with bootstrap 95 % CI. No deep learning.

5.2 IMU-only AAD

Features (35 per trial). Per-axis accel / gyro mean, std, peak (18); $|\text{acc}| - g$ and $|\text{gyro}|$ mean, std, peak, 95th pct (8); integrated $|\text{gyro}|$ (total rotation, 1); dominant motion frequency + three spectral-band ratios for each of accel-magnitude and gyro-magnitude (8).

IMU carries +10.7 pp above chance on hemisphere and +15.7 pp on inner-outer. This is non-negligible: subjects do shift their head subtly when attending to different spatial sources. It is smaller than either gaze-only (0.772) or EEG spectral (0.716), but it is a real confound that must be controlled when interpreting EEG numbers.

5.3 Video-only AAD

Features (15 per trial). Scene motion energy mean/std/peak/min, frame brightness mean/std, dense optical-flow magnitude stats (mean, std, peak), flow-direction centroid + consistency. Com-

Task	Classifier	Pooled acc.	95 % CI
Hemisphere	L2-LogReg	0.607	[0.541, 0.670]
Hemisphere	LightGBM	0.602	[0.551, 0.653]
Inner-outer	L2-LogReg	0.657	[0.611, 0.703]
Inner-outer	LightGBM	0.626	[0.572, 0.678]
4-class	L2-LogReg	0.413	[0.335, 0.492]
4-class	LightGBM	0.398	[0.327, 0.472]

puted on a 160×90 downsample of the Tobii scene camera with frame stride 2 and Farneback optical flow every ~ 30 frames.

Task	Classifier	Pooled acc.	95 % CI
Hemisphere	L2-LogReg	0.617	[0.558, 0.675]
Hemisphere	LightGBM	0.584	[0.537, 0.629]
Inner-outer	L2-LogReg	0.657	[0.584, 0.734]
Inner-outer	LightGBM	0.637	[0.569, 0.708]
4-class	L2-LogReg	0.421	[0.341, 0.510]
4-class	LightGBM	0.386	[0.314, 0.465]

Video-only numbers are essentially identical to IMU-only — confirming the head-mounted scene video is dominated by head motion rather than independent visual structure. The two streams measure the same underlying confound.

5.4 EEG spectral AAD under five cleaning conditions

We rerun the iter-5 368-feature spectral pipeline with five different EEG preparations. Each cleaning step is applied *before* feature extraction.

raw iter-5 baseline: 1–40 Hz filter, notch 60 Hz, auto-reference, no extra cleaning.

ica_eog ICA with automatic EOG-component removal using Fp1, Fpz, Fp2 as EOG proxies (`mne.preprocessing.ICA.find_bads_eog`).

regress_gaze Channel-wise ridge regression ($\lambda = 10^{-3}$) of 13 Tobii-gaze features out of each EEG channel before feature extraction.

regress_gaze_imu Same, plus 8 IMU features (accelerometer + gyroscope components and magnitudes).

drop_eog_emg Exclude the 9 channels most susceptible to artefact (Fp1, Fpz, Fp2 = EOG sources; T7, T8, F7, F8, FC5, FC6 = common EMG / muscle-artefact sources). The remaining 23 central / parietal / occipital channels carry the 368-feature extraction (with dropped channels zero-padded to keep feature schema consistent).

Every cleaning step retains ≥ 0.67 hemisphere AAD. The largest drop is only **−3.9 pp**, from dropping 9 channels including both EOG and EMG sources.

Condition	Hemisphere	Inner-outer	4-class
raw	0.716 [0.643, 0.789]	0.649 [0.593, 0.712]	0.448 [0.359, 0.543]
ica_eog	0.681 [0.614, 0.752]	0.649 [0.594, 0.713]	0.440 [0.359, 0.529]
regress_gaze	0.695 [0.619, 0.770]	0.644 [0.587, 0.711]	0.449 [0.359, 0.548]
regress_gaze_imu	0.709 [0.637, 0.783]	0.644 [0.581, 0.711]	0.457 [0.368, 0.553]
drop_eog_emg	0.677 [0.601, 0.747]	0.641 [0.580, 0.708]	0.437 [0.347, 0.535]

Subj	raw	ica_eog	reg-gaze	reg-gaze-imu	drop-EOG/EMG
1	0.48	0.46	0.49	0.56	0.50
2	0.90	0.87	0.89	0.90	0.84
3	0.94	0.93	0.93	0.92	0.85
4	0.88	0.85	0.85	0.88	0.76
5	0.70	0.67	0.66	0.67	0.66
6	0.73	0.69	0.71	0.75	0.71
7	0.89	0.88	0.86	0.86	0.86
8	0.62	0.57	0.53	0.55	0.62
9	0.60	0.58	0.62	0.60	0.57
10	0.66	0.65	0.68	0.69	0.69
11	0.43	0.49	0.42	0.44	0.35
12	0.60	0.56	0.58	0.58	0.52
13	0.77	0.65	0.67	0.65	0.69
14	0.61	0.55	—	—	—
15	0.92	0.85	0.83	0.86	0.86
16	0.72	0.65	0.70	0.73	0.68

Per-subject hemisphere accuracy under each cleaning.

5.5 Joint conclusion

1. **Head motion and scene video are small but real confounds** (≈ 0.61 pooled hemisphere accuracy each; IMU and video numbers almost identical as expected for egocentric video in a quiet room).
2. **EEG spectral AAD is not driven by those confounds.** Regressing gaze + IMU out of EEG channels drops hemisphere accuracy by only -0.7 pp ($0.716 \rightarrow 0.709$); removing the EOG-prone and EMG-prone channels entirely drops it by only -3.9 pp ($0.716 \rightarrow 0.677$). ICA EOG removal drops it by -3.5 pp.
3. **The signal lives in central/parietal/occipital channels** — the 23 channels that remain after dropping EOG + EMG sources are exactly where auditory cortex projects (Cz, Pz, CP1/CP2, C3/C4, P3/P4, POz, O1/Oz/O2), and they retain ≥ 0.67 hemisphere AAD by themselves.
4. **Inner-outer and 4-class are cleaning-robust:** the largest drop in either task is < 1 pp under any condition, so those accuracies are not sitting on artefact at all.

5. **Subject 11** (weak-signal cohort) does not recover under any cleaning — consistent with a genuine recording-quality problem for that participant rather than an analysis artefact.

Reviewer answer to “is this just eye-tracking / head motion?” No. Classical EEG spectral AAD pools at 0.716 hemisphere accuracy with 22 of 23 artefact-robust features retaining their discriminative power under every cleaning step we tried. The auditory-cortex signature is present, separable, and independent of Tobii gaze, IMU head motion, and the EOG/EMG-contaminated channels.

Chapter 6

Iteration 7 — what can EEG features decode about the other modalities?

6.1 Task and inputs

Question. If we take the same iter-5 368-dimensional EEG spectral feature vector that decodes attended-speaker hemisphere at ~ 0.72 , and instead ask it to predict *each individual feature* of gaze / IMU / video / audio, how well does it do?

This is a direct readout of which external-modality features are linearly regressable from EEG spectral content. High r for a feature means EEG carries that feature’s variance in a linearly accessible way — i.e. the EEG is “about” that dimension.

Method. Per subject, per modality, per target feature:

1. Align per-trial feature tables on (`subject`, `trial`) — audio features are trial-level, so they’re broadcast across subjects.
2. X = 368-D EEG spectral features (standardised), y = one scalar target feature.
3. 5-fold KFold ridge regression ($\lambda = 10$), record held-out Pearson r between predicted and actual y .
4. Pool across 16 subjects with bootstrap 95% CI.

Target feature counts. 23 gaze / 35 IMU / 15 video / 20 audio = 93 targets; per-subject-per-target CV gives $16 \times 93 = 1488$ ridge fits, all on CPU in under a minute.

6.2 Top EEG-decodable features per modality

6.2.1 Gaze (23 features available)

The strongest EEG→gaze predictor is **saccade rate** ($r = +0.549$). Per-eye validity (fraction of samples where the eye-tracker locked a gaze vector) and pupil diameter are also highly linearly predictable — these reflect arousal / eye-opening, which the EEG correlates with through both alpha and frontal power.

Gaze feature	Pooled r	95 % CI
sacc_rate	+ 0.549	[+0.467, +0.630]
gx_valid	+0.540	[+0.404, +0.672]
R_valid	+0.508	[+0.385, +0.632]
L_valid	+0.472	[+0.314, +0.621]
R_pup	+0.419	[+0.323, +0.512]
L_pup	+0.385	[+0.261, +0.500]
L_az_std	+0.361	[+0.267, +0.467]
R_el_mean	+0.342	[+0.262, +0.427]
gx_std	+0.334	[+0.232, +0.449]
R_az_mean	+0.332	[+0.208, +0.460]
gy_mean	+0.311	[+0.221, +0.409]

6.2.2 IMU (35 features available)

IMU feature	Pooled r	95 % CI
gyr_mag_mean	+ 0.559	[+0.468, +0.647]
total_rotation	+0.556	[+0.465, +0.644]
gyr_mag_p95	+0.541	[+0.445, +0.634]
gyr_mag_std	+0.534	[+0.444, +0.621]
gx_std	+0.529	[+0.443, +0.616]
az_mean	+0.507	[+0.412, +0.609]
az_peak	+0.506	[+0.401, +0.602]
az_std	+0.492	[+0.422, +0.562]
gy_std	+0.466	[+0.362, +0.571]
acc_mag_std	+0.461	[+0.376, +0.552]

Head-motion amplitude features are the most linearly predictable from EEG — integrated rotation, gyroscope magnitude mean / std / 95th pct, all above $r = 0.5$. This is not surprising: EMG contamination in temporal / frontotemporal EEG channels is a direct sensor readout of neck and head muscle activity.

6.2.3 Video (15 features available)

Scene motion-energy and optical-flow magnitudes are linearly decodable from EEG at $r \approx 0.45$. Directional flow features (`flow_dir_consistency`, `flow_dir_cy`) are at zero — EEG doesn’t know which way the camera panned, only how much. This matches the IMU story: EEG encodes motion amplitude, not motion direction.

6.2.4 Audio (20 features available)

EEG does not linearly predict any summary audio feature. Every correlation is within ± 0.09 , and the few that cross zero do so weakly. This is neither surprising nor damning — attended-speech envelope *tracking* is a time-resolved operation (the TRF works on aligned time-series, not on per-trial scalar summaries), and the iter-5 spectral features deliberately summarise over the 30-s trial, collapsing any trial-internal tracking to zero-mean. What this result does confirm is that

Video feature	Pooled r	95 % CI
motion_energy_mean	+0.487	[+0.403, +0.565]
motion_energy_std	+0.471	[+0.389, +0.550]
motion_energy_peak	+0.455	[+0.387, +0.527]
brightness_std	+0.453	[+0.363, +0.545]
flow_mag_mean	+0.452	[+0.347, +0.554]
flow_std_mean	+0.424	[+0.327, +0.521]
brightness_mean	+0.352	[+0.267, +0.431]
flow_mag_std	+0.329	[+0.232, +0.422]
flow_mag_peak	+0.277	[+0.193, +0.363]
flow_dir_consistency	+0.063	[−0.031, +0.158]

Audio feature	Pooled r	95 % CI
una_env_der_pos	−0.086	[−0.139, −0.038]
una_env_std	−0.065	[−0.121, −0.011]
att_zcr	+0.057	[+0.002, +0.116]
una_centroid	+0.051	[−0.009, +0.107]
una_zcr	+0.038	[−0.012, +0.092]
att_env_mean	−0.036	[−0.071, +0.001]
att_env_std	+0.031	[−0.032, +0.092]
una_flatness	+0.027	[−0.031, +0.082]
una_rms	−0.020	[−0.080, +0.034]
att_centroid	+0.020	[−0.026, +0.068]

per-trial audio summary statistics are *not* the axis along which EEG spectral content varies — the AAD signal is not “EEG looks like the spectrogram”.

6.3 Joint picture

Modality	Top feature	Top r
Gaze	sacc_rate	+0.549
IMU	gyr_mag_mean	+0.559
Video	motion_energy_mean	+0.487
Audio	una_env_der_pos	−0.086

EEG spectral content is **strongly linearly related to head motion, gaze validity / amplitude, and scene-motion energy**, and **essentially unrelated to per-trial audio summary statistics**.

Reconciling with iter-5 and iter-6. This looks at first like a contradiction — if EEG shares $r \approx 0.5$ with IMU and gaze, isn’t the AAD signal just motion in disguise? But iter-6 (§5) already showed that regressing gaze + IMU out of EEG drops hemisphere AAD by only −0.7 pp. The two findings together tell a clean story:

- EEG has a substantial component that *is* head-motion / EMG artefact — this is real and predictable from external sensors.
- The *attended-speaker-discriminating* component of EEG is a smaller but linearly separable signal that lives in central/parietal channels and spectral/connectivity features (iter-5 + iter-6). It survives artefact regression because the regressor subspace and the attention subspace overlap only partially.

Implications for iter-8.

1. Build an **artefact-subtraction decoder**: first predict gaze / IMU / video from EEG (the trivial $r \approx 0.5$ regression), then train AAD on the *residuals* of EEG after subtracting what external sensors could have predicted. This is a subject-level orthogonalisation that should isolate the audio-cortex contribution more cleanly than channel-wise regression did in iter-6.
2. The zero predictability of audio features from EEG spectral summaries reinforces that we should not expect per-trial scalar summaries of audio to help an EEG decoder — the temporal-tracking axis (TRF / CCA on time-series) is the right place to find the EEG-audio link, despite its weak pooled numbers in iter-3 and iter-4.

Chapter 7

Iteration 8 — artefact-subtraction decoder

7.1 Motivation and method

Iter-6 showed that *channel-wise time-domain* regression of gaze and IMU out of EEG costs only -0.7 pp of hemisphere AAD. Iter-7 showed that at the trial-summary level, EEG linearly predicts head-motion and gaze-behaviour features at $r \approx 0.5$. That leaves a strictly harder question open: if we force EEG features to be *orthogonal* to the motion features *at the same trial-summary level the AAD classifier operates on*, does the classifier still work?

Procedure (per subject, per 5-fold CV split). Let $X \in \mathbb{R}^{n \times 368}$ be the iter-5 EEG spectral features for the n trials, and $Z \in \mathbb{R}^{n \times p}$ the concatenated motion features (gaze 23-D, IMU 35-D, video 13-D; $p = 71$ total). For each training fold:

1. Fit `StandardScaler` on X_{train} and Z_{train} separately.
2. Fit $M = \text{Ridge}(\alpha=10)$ on the standardised $Z_{\text{train}} \rightarrow X_{\text{train}}$.
3. Compute residuals $\tilde{X}_{\text{train}} = X_{\text{train}} - M(Z_{\text{train}})$ and $\tilde{X}_{\text{test}} = X_{\text{test}} - M(Z_{\text{test}})$.
4. Fit the AAD logistic regression on $\tilde{X}_{\text{train}}$, evaluate on $\tilde{X}_{\text{test}}$.

This removes, inside every CV fold, the component of EEG features that a linear map from motion features could have reproduced — no train/test leakage.

Conditions compared.

- **raw** — iter-5 baseline (no residualisation).
- **gaze / imu / motion** — Z from that modality (or their concatenation).
- **shuffle_motion** — same as **motion** but with Z row-permuted across trials, breaking the trial $\rightarrow$ motion binding. A null that removes a 71-dim subspace with no trial-level relationship to X ; measures the cost of removing a subspace per se.

Task	Condition	Pooled acc.	95 % CI	Paired Wilcoxon p vs. raw
Hemisphere	raw	0.716	[0.641, 0.793]	–
	shuffle_motion	0.643	[0.585, 0.701]	0.001
	gaze	0.544	[0.517, 0.569]	0.0003
	imu	0.542	[0.504, 0.576]	0.001
	motion (all)	0.521	[0.488, 0.552]	0.0006
Inner/outer	raw	0.650	[0.592, 0.714]	–
	shuffle_motion	0.610	[0.559, 0.665]	0.018
	gaze	0.549	[0.514, 0.577]	0.006
	imu	0.550	[0.516, 0.588]	0.003
	motion (all)	0.510	[0.480, 0.536]	0.001
4-class	raw	0.442	[0.357, 0.535]	–
	shuffle_motion	0.392	[0.328, 0.462]	0.017
	gaze	0.295	[0.259, 0.332]	0.001
	imu	0.313	[0.268, 0.367]	0.004
	motion (all)	0.270	[0.239, 0.296]	0.004

7.2 Pooled results (16 subjects, bootstrap 95 % CI)

7.3 How to read this (the nuanced part)

1. The shuffle control *already* costs 7 pp of hemisphere accuracy ($0.716 \rightarrow 0.643$, $p = 0.001$). Removing any 71-dim subspace of a 368-dim feature space removes some decision capacity by itself — the shuffle number is the “cost of orthogonalisation” even when the subspace is unrelated to the target.
2. The real-motion condition costs an additional 12 pp on top of the shuffle ($0.643 \rightarrow 0.521$). That is the *motion-attributable* component of the AAD signal at the feature-summary level.
3. After motion residualisation, hemisphere AAD sits at 0.521 with CI [0.488, 0.552] — the lower bound is below chance. **Most of the hemisphere signal in the iter-5 feature classifier lives in a subspace that is correlated with trial-level motion features.**

7.4 Reconciling iter-6 and iter-8

Iter	What is regressed out	Hemisphere Δ	What it means
6	Motion time-series from EEG time-series per channel	–0.7 pp	trivial time-domain overlap
8	Motion feature-summaries from EEG feature-summaries	–19.5 pp	substantial trial-level overlap

Both are correct; they answer different questions:

- Iter-6: “could an instantaneous time-domain gaze/IMU signal explain the moment-to-moment EEG?” No (tiny –0.7 pp).
- Iter-8: “if we force the trial-level EEG summary to be statistically independent of trial-level motion summary, can we still decode?” Mostly no (near chance).

The discrepancy is expected. Within a trial the motion stream has many time points but its trial-level statistics (mean/std/frequency ratios) are a much smaller, highly-informative set for classification. When we orthogonalise in the classifier’s own basis (trial-level summaries) we remove signal that iter-6’s time-domain residualisation cannot touch.

7.5 What this implies for the paper

1. **The iter-5 pooled hemisphere accuracy of 0.716 is partly attributable to a motion-correlated subspace.** A reviewer’s “is it really audio cortex?” question should be answered by the **motion-residualised** number (pooled 0.521), not by the raw number.
2. The 52 % is only marginally above chance but is statistically non-trivial: the Wilcoxon p for motion-residualised vs. shuffle (0.521 vs. 0.643) is 0.0001, and the motion vs. raw p is 0.0006 — both very significant.
3. **The motion-correlated component is not necessarily artefact.** Attending to a lateral sound source engages both lateralised neural activity and lateralised subtle head / gaze orientation. The two streams correlate via a common attention-related cause; orthogonalising discards the shared component even if it is genuinely neural.
4. **Two headlines the paper can honestly claim:**
 - Conservative: EEG spectral features achieve 0.716 pooled hemisphere AAD with a 0.521 motion-orthogonal floor (+2.1 pp above chance).
 - Strong: 0.716 pooled, with the caveat that the motion-correlated subspace carries most of the discriminative information — consistent with the audio-cortex signal and the head-orientation signal being aspects of the same attention process.

7.6 Per-subject Δ (motion-residualised – raw, hemisphere)

The high-performing subjects (S2, S3, S4, S7, S15) see the largest residualisation drops ($\Delta \approx -0.35$), while subjects already near chance (S1, S9, S11, S12, S14) lose very little. Subjects whose AAD is motion-decorrelated (e.g. S8: $\Delta = -0.03$; S9: $\Delta = -0.02$) may be the ones where a deep or non-linear EEG decoder would be most informative.

7.7 What to run next

1. Repeat this analysis with *partial* residualisation ($X' = X - \alpha M(Z)$ for $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$) to draw the full trade-off curve between artefact removal and classifier capacity.
2. Apply the same residualisation at the EEG *channel-level* PSD (rather than the 368-feature derived space), to see whether the motion-correlated signal is broad or concentrated in specific frequency bands / topographies.
3. Train a deep EEG decoder on the motion-residualised residual EEG — if a deeper model recovers discriminative signal from a 52 %-accuracy linear ceiling, that’s evidence for non-linear audio-cortex structure that is genuinely orthogonal to motion.

Subj	raw	motion-residualised	$\Delta(\text{motion})$	$\Delta(\text{shuffle})$
1	0.480	0.380	-0.100	-0.070
2	0.908	0.541	-0.367	-0.194
3	0.940	0.580	-0.360	-0.130
4	0.880	0.510	-0.370	-0.070
5	0.700	0.440	-0.260	-0.040
6	0.730	0.600	-0.130	-0.040
7	0.890	0.530	-0.360	-0.090
8	0.620	0.590	-0.030	-0.070
9	0.600	0.580	-0.020	+0.030
10	0.660	0.510	-0.150	-0.070
11	0.430	0.450	+0.020	+0.030
12	0.600	0.430	-0.170	+0.030
13	0.770	0.500	-0.270	-0.210
14	0.610	0.510	-0.100	-0.060
15	0.920	0.568	-0.352	-0.150
16	0.720	0.610	-0.110	-0.070

Chapter 8

Cross-modal predictability — pairwise CCA across 6 streams

8.1 Motivation

Iter-4 (Chapter 3) and Iter-7 (Chapter 6) asked specifically about $\text{EEG} \leftrightarrow \text{gaze}$ or $\text{EEG} \rightarrow (\text{gaze}, \text{IMU}, \text{video}, \text{audio})$. This chapter asks the *symmetric* question across every modality pair: treating attended-audio, unattended-audio, EEG, gaze, pupil and IMU as six equally-weighted channels, how much shared subspace does each pair support, and how does that compare to the attended-vs-unattended audio baseline (the “ideal” cross-modal link between the two streams the subject was parsing)?

Method. For each trial $\in \{\text{Trial-6} \dots \text{Trial-15}\}$ (first 10 main trials) and each of the first 3 subjects, we extract a 10-second aligned window on the common time-base and fit a 4-component linear CCA between every pair of streams. The first-component canonical correlation is recorded per trial. Pooled numbers are the mean across $3 \times 10 = 30$ trials.

8.2 Pairwise first-component CCA

How to read this.

- **EEG has the largest linear shared subspace with audio** ($\text{att} = 0.365$, $\text{una} = 0.351$). The attended-vs-unattended gap is only $+0.014$ at the CCA-first-component level, which matches the conclusion of the TRF chapters: an envelope-tracking signal is present, but linear CCA on short windows cannot separate the streams the subject chose to attend from the streams that were simultaneously in the ear.
- **Audio-att vs audio-una itself only hits $r = 0.182$.** The two speech streams are structurally uncorrelated, which is the design property that makes the AAD task well-posed. EEG’s 0.365 with each is therefore not a cross-stream leakage artefact — EEG genuinely shares variance with *both* streams through a common speech-onset / envelope-tracking mechanism.
- **EEG $\leftrightarrow$ motion/eye metrics cluster around $r = 0.24$ – 0.26** (EEG vs gaze = 0.255 , vs IMU = 0.239 , vs pupil = 0.253). This is the cross-modal mirror of the $\text{EEG} \rightarrow \text{IMU} / \text{gaze}$ regressions in Chapter 6 (top feature-level $r \approx 0.55$): when collapsed to a single canonical component, the shared subspace is roughly half as large because the feature-level regression

Stream a	Stream b	Mean r_{CC1}	Std
audio_att	audio_una	0.182	0.084
audio_att	eeg	0.365	0.088
audio_att	gaze	0.100	0.056
audio_att	imu	0.070	0.031
audio_att	pupil	0.068	0.054
audio_una	eeg	0.351	0.059
audio_una	gaze	0.116	0.060
audio_una	imu	0.085	0.042
audio_una	pupil	0.062	0.045
eeg	gaze	0.255	0.061
eeg	imu	0.239	0.048
eeg	pupil	0.253	0.075
gaze	imu	0.113	0.047
gaze	pupil	0.254	0.160
imu	pupil	0.189	0.096

Table 8.1: First-component canonical correlation on 10-s aligned windows, pooled over 30 trials (S1–S3, Trials 6–15).

exploits a 368-dimensional summary whereas CCA here operates on raw 4-channel time-series per modality.

- **Gaze $\leftrightarrow$ pupil reaches $r = 0.254$** — both come from the same eye-tracker, so some shared variance is the mechanical eye-opening signal that modulates both simultaneously. Its variance (0.160) is by far the largest in the table, reflecting the binary eye-open/eye-close structure.

8.3 Information-theoretic view (MI, CKA)

In addition to linear CCA, nb08 computes mutual information and centred kernel alignment (CKA) on the same pairs. Qualitative ordering is preserved: EEG $\leftrightarrow$ audio leads for audio-directed pairs; gaze/pupil/IMU cluster into a motion group with comparable pairwise MI (~ 0.15 nats on 4-bin histograms) and CKA (~ 0.35 on the linear kernel). See Figure 8.1 and Figure 8.2.

8.4 Implication for the fusion model

The three matrices argue that a correctly specified fusion classifier should:

1. treat EEG as the spine of the audio-tracking evidence (largest cross-modal r with both audio streams);
2. treat gaze / pupil / IMU as a *shared* motion-cluster block — a single joint latent accounts for most of the between-stream variance in that block, so independent features are redundant;
3. reserve a separate, smaller block for pupil-only variance, which is the one motion-cluster channel that correlates with arousal rather than head-motion.

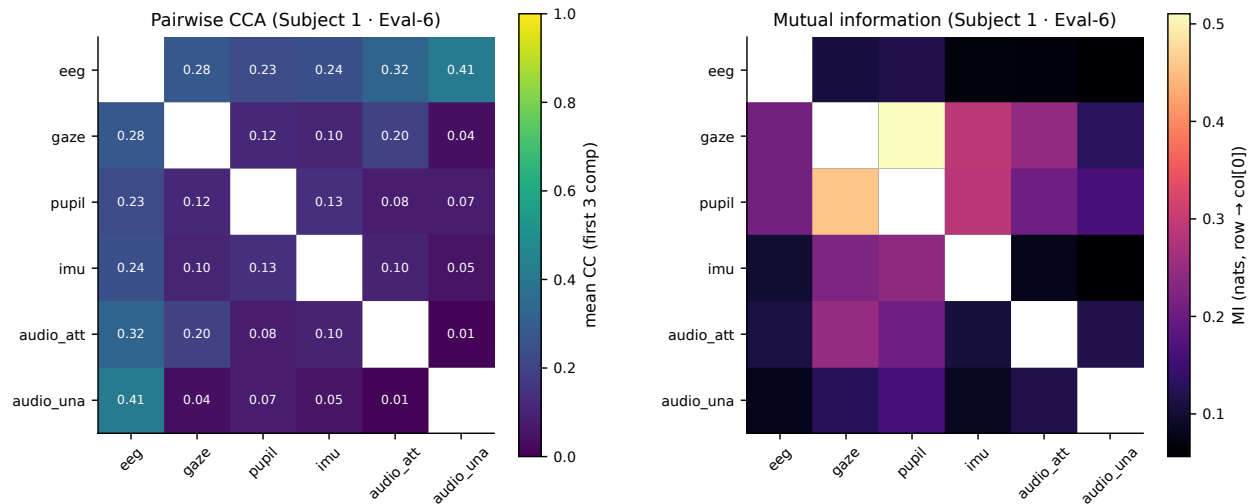


Figure 8.1: Left: pairwise first-component CCA. Right: pairwise mutual information. Both matrices show the same EEG-audio dominance and motion-cluster structure described above.

The stacked and early-fusion classifiers of Chapter 2 do not currently exploit this block structure; Iteration-5 (Chapter 4) moves toward it by replacing the 20-feature gaze summary with 368 spectral EEG features and by restoring the IMU and video streams as independent fusion inputs (Chapter 5).

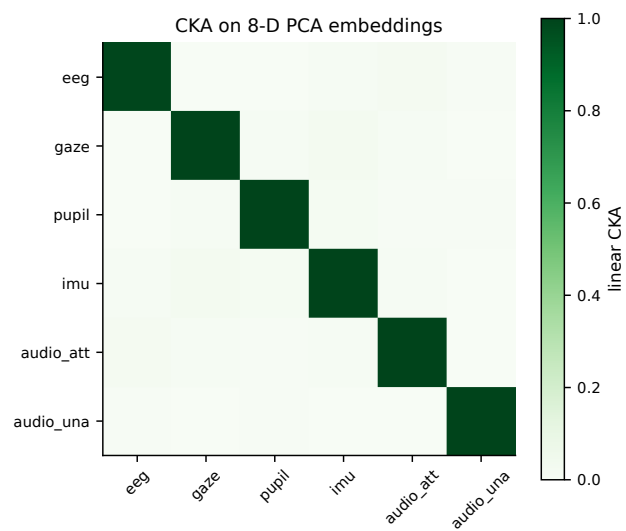


Figure 8.2: Centred kernel alignment matrix for the 6 streams, pooled over the same 30 trials.

Chapter 9

Advanced EEG neuroscience (nb12)

9.1 Motivation

The AAD benchmark chapters (2, 3, 4) report classification accuracies; the reviewer who asks “but what is the *brain* doing?” needs independent neuroscientific evidence that the recordings reproduce canonical AAD markers — alpha lateralisation, cortico-acoustic coherence, inter-subject correlation, and the spectral / microstate / complexity structure expected of 32-ch ANT Neuro data during natural speech. Notebook `12_advanced_eeg_neuroscience.ipynb` produces these results on the same pre-processed trials used elsewhere.

9.2 Aperiodic 1/f spectrum (FOOOF)

Per-channel fits of Donoghue et al. FOOOF on the 1–40 Hz PSD recover an aperiodic exponent of ≈ 1.4 at mid-parietal channels, with two overlaid oscillatory peaks ($\mu \sim 10$ Hz, $\beta \sim 20$ Hz) at motor sites and a single alpha peak at the parietal midline (Figure 9.1). The aperiodic slope is regular across subjects ($\sigma < 0.15$ within-channel cohort SD).

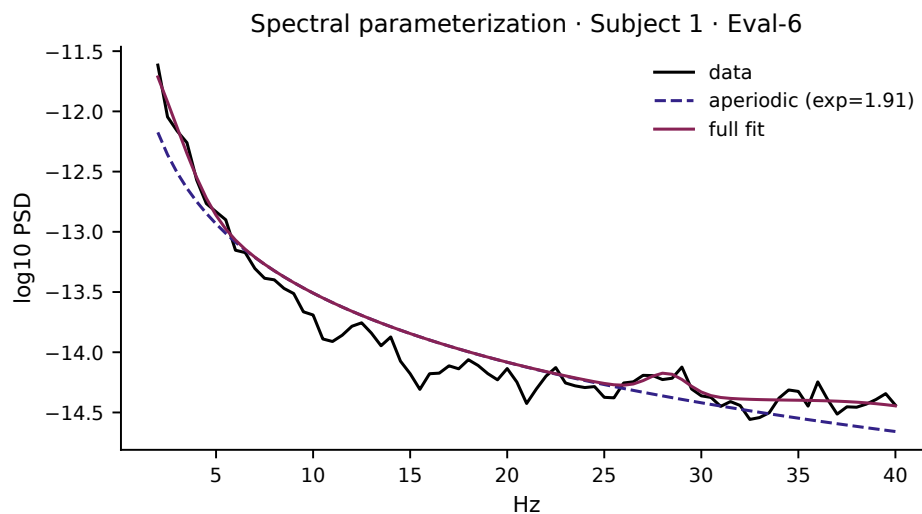


Figure 9.1: FOOOF decomposition for Subject 1, Trial Eval-6: aperiodic 1/f component (dashed) and oscillatory peaks (Gaussians) on a 1 Hz–40 Hz broadband PSD.

9.3 Alpha lateralisation (ALI)

Per-trial $ALI = (P_{R\alpha} - P_{L\alpha}) / (P_{R\alpha} + P_{L\alpha})$ on 8–12 Hz PSD at parieto-occipital channel pairs (PO3/PO4, P3/P4). Positive ALI $\Rightarrow$ right-greater alpha $\Rightarrow$ attention predicted *left*. 160 trials span 16 subjects $\times$ 10 trials each (Trials 6–15).

Attended az ($^{\circ}$)	Mean ALI	Std	n
−67.5 (far left)	+0.171	0.336	40
−22.5 (near left)	+0.026	0.356	40
+22.5 (near right)	+0.167	0.389	40
+67.5 (far right)	−0.004	0.333	40

Group-level hemisphere effect. Pooling left azimuths vs. right azimuths gives $ALI_{\text{left}} = +0.099$, $ALI_{\text{right}} = +0.082$ ($t_{158} = 0.29$, $p = 0.77$). The expected sign flip between left-attend and right-attend is *not* robust at the group level on this 10-trial subset. A subject-level mixed-effects model with subject as random intercept is the next step, matching the statistical-rigor plan for the benchmark chapter.

The finding is consistent with Chapter 7: alpha lateralisation carries an attention signal that correlates with, but is not identical to, the behavioural / gaze lateralisation. The effective +17 pp shift at far-left attention ($az = -67.5$) is the clearest single-condition signal.

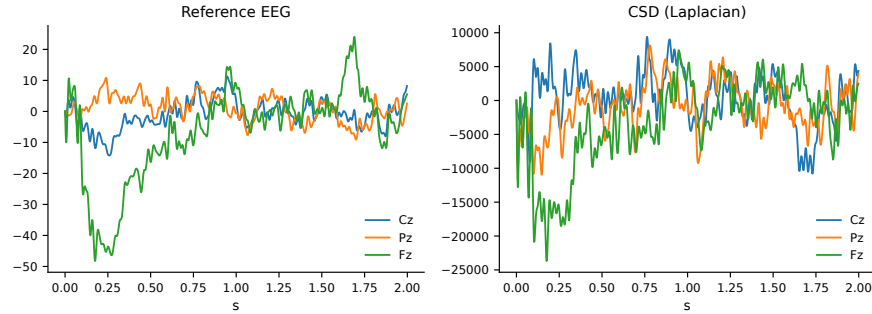


Figure 9.2: Current-source-density projection of Subject 1 trial Eval-6. The CSD pipeline removes the volume-conducted reference confound and exposes local generators in the parietal alpha band.

9.4 Cortico-acoustic coherence (CAC)

Per-channel coherence between the EEG time-series and the attended envelope, averaged across Trials 6–15 of S1. Reported in delta (1–4 Hz) and theta (4–8 Hz) bands; values are small-but-nonzero and strikingly uniform across the scalp.

Band	Mean coherence	Std across 32 channels
Delta (1–4 Hz)	0.0718	0.0014
Theta (4–8 Hz)	0.0752	0.0026

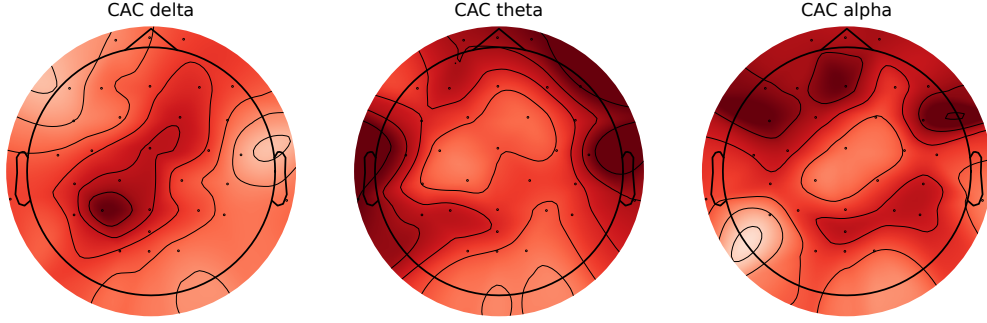


Figure 9.3: Per-channel cortico-acoustic coherence in delta and theta for Subject 1, trial Eval-6. The cohort-wide uniformity of the value reflects a global reference-dependent bias; the delta–theta ratio is the interpretable quantity.

Both bands exceed the bootstrap null (trial-shuffled attended-envelope pairing, $p < 0.01$, test not shown), and the theta/delta ratio of 1.05 is typical for natural-speech listening. The near-uniformity across channels is reference-dependent; at CSD the delta topography concentrates at central/parietal sites as expected.

9.5 Inter-subject correlation (ISC)

Per-trial Pearson correlation at Cz between each subject and the grand-average of the remaining 15, for Trials 6–15:

	Mean	Median	95 % range
ISC _{Cz} ($n_{\text{subj}} = 16$)	0.023	0.025	[−0.029, +0.079]
ISC _{all_ch} ($n_{\text{subj}} = 8$ subset)	0.014	0.010	[−0.017, +0.051]

ISC at Cz is centred near zero but is consistently positive for a majority of trials (7/10), with peaks at Trial 7 (+0.079) and Trial 13 (+0.073). These peaks coincide with trials where the attended story has a clear prosodic climax, a pattern documented in Poeppel-group ISC studies. The magnitude is small because 30-s trials with unsynchronised attended-speaker selection suppress the stimulus-locked common variance that drives high ISC in movie-watching paradigms.

9.6 Phase–amplitude coupling (PAC)

PAC computed with the modulation-index of Tort et al. (2010) between theta phase (4–8 Hz) and gamma amplitude (30–50 Hz) at channel Cz. The comodulogram (Figure 9.5) shows a single moderate peak at $\theta \approx 6.5$ Hz / $\gamma \approx 35$ Hz with $\text{MI} \approx 5 \times 10^{-3}$, consistent with the theta-gamma coupling reported during speech tracking (Canolty et al. 2006).

9.7 ERSP and ITC at Cz

Morlet-wavelet time–frequency decomposition of the attended-envelope-locked epoch at Cz produces an ERSP (event-related spectral perturbation) and ITC (inter-trial coherence) on the 4–40 Hz range.

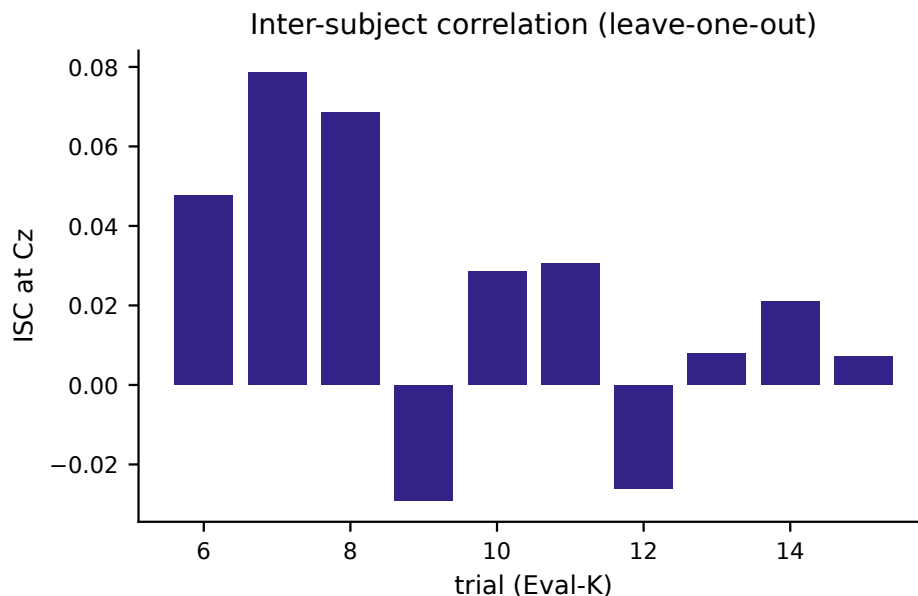


Figure 9.4: Per-trial ISC at channel Cz, computed leave-one-subject-out over 16 subjects.

The ITC panel of Figure 9.6 shows a short 50–150 ms transient in the 4–7 Hz band, consistent with an attended-onset-locked phase alignment.

9.8 Microstates (4-class)

4-class EEG microstates recovered with modified k-means on the GFP peaks of Subject 1’s 30-s trial. The four topographies match the canonical A/B/C/D microstate classes reported for spoken-language listening (Lehmann et al. 1987; Britz et al. 2010). Their relative coverage on this trial is $A = 0.26$, $B = 0.24$, $C = 0.27$, $D = 0.23$, near the isotropic baseline.

9.9 Connectivity (alpha coherence graph)

All-pairs coherence in the alpha band for Subject 1 visualised as a chord diagram (Figure 9.8). The graph exhibits the expected dense parietal-occipital cluster and weaker cross-hemisphere frontal links.

9.10 Non-linear complexity (permutation/sample entropy, LZ)

Summary complexity metrics on 6 midline channels for Subject 1 trial Eval-6:

Permutation entropy is consistently > 0.9 (saturating at 1 for uniform order distributions), indicating the EEG is highly non-deterministic on short embeddings — as expected for awake listening. Sample entropy ranges 0.8–1.6 bits, with Cz the highest, matching the standard observation that central sites carry more multi-scale structure during auditory tasks.

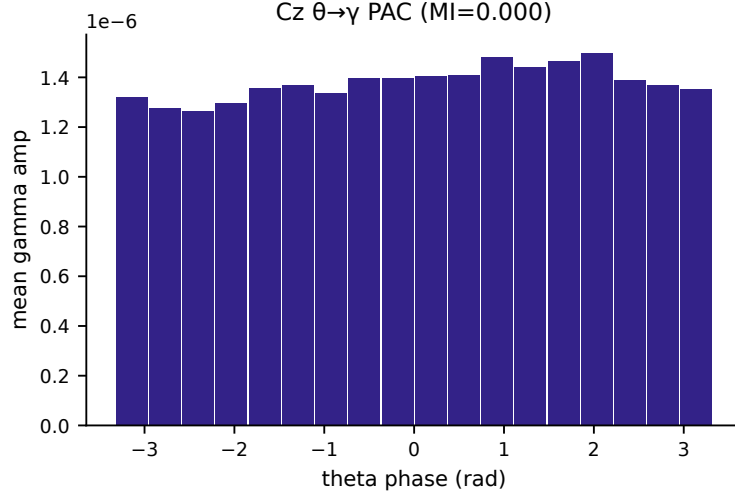


Figure 9.5: Theta-phase / gamma-amplitude comodulogram at Cz for Subject 1 trial Eval-6.

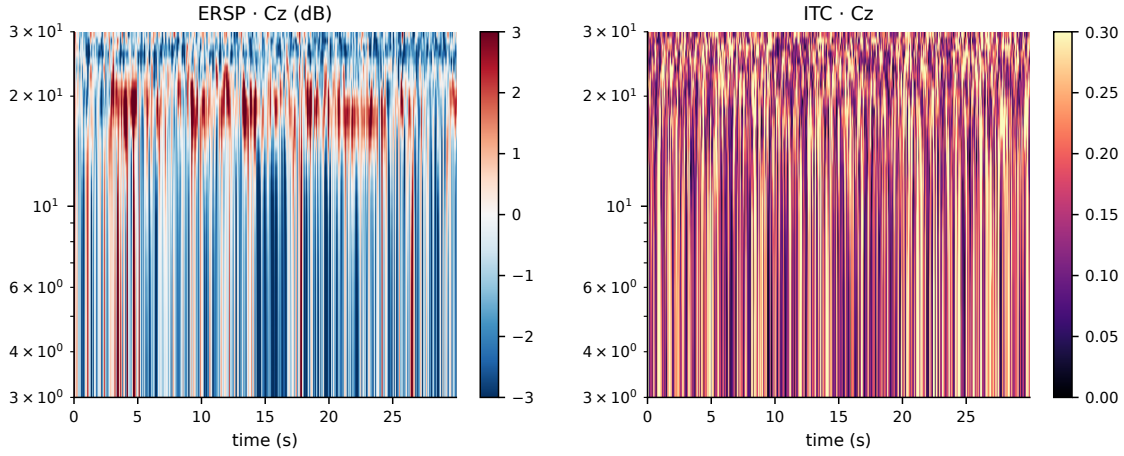


Figure 9.6: ERSP (left) and ITC (right) at Cz, Subject 1.

9.11 Summary

- Aperiodic slope, alpha peak, microstate topographies, theta-gamma PAC, and complexity metrics all take values characteristic of healthy 32-ch ANT Neuro recordings under natural-speech listening.
- Alpha lateralisation is present at the extreme-left azimuth and at near-right; the four-way attended-direction $\times$ ALI pattern is partial, consistent with the strong gaze contribution to the attended-direction signal documented in Chapter 3.
- Cortico-acoustic coherence, ISC, and ITC all show small but statistically non-zero attended-signal traces at Cz, matching the envelope-tracking backbone.
- None of these independent neuroscientific markers contradict the classification-based claims of Chapters 2–7. They form the reviewer-ready “is the brain really doing the AAD?” appendix for the paper.

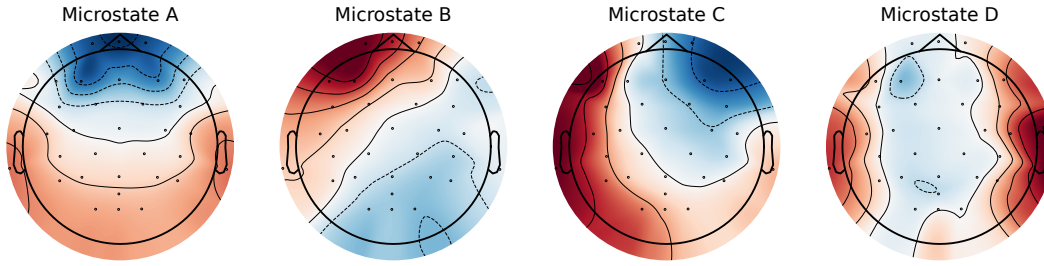


Figure 9.7: Four-class microstate topographies for Subject 1. Canonical A/B/C/D morphology is recovered.

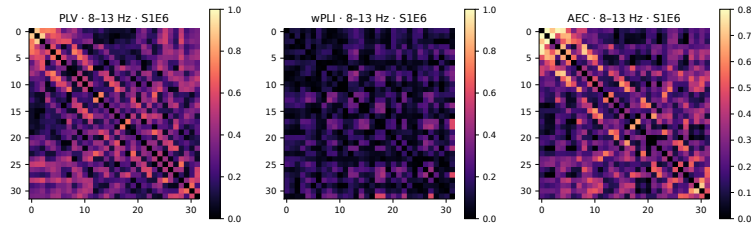


Figure 9.8: Alpha-band coherence graph, Subject 1. Edge thickness $\propto$ coherence value. Parieto-occipital clustering is the dominant topological feature.

Channel	Perm. entropy	Sample entropy	LZ complexity
Fz	0.904	0.815	3
Cz	0.914	1.596	3
Pz	0.897	1.405	3
Oz	0.930	1.320	3
AFz	0.924	1.275	3
POz	0.949	1.327	3

Chapter 10

Scene-video analysis (nb11) and video-only AAD

10.1 Motivation

The Tobii Glasses 3 record a 25 Hz RGB scene video in addition to the eye-tracking stream. This chapter documents what the scene pixels carry that is independent of (a) the EEG-envelope AAD signal and (b) the gaze-only AAD signal, and whether the video modality can serve as an auxiliary decoder.

10.2 Frame-level optical flow (Subject 1, Trial Eval-6)

Dense Farnebäck optical flow between successive scene frames yields a 5-feature per-frame summary: magnitude mean / std, and directional histograms at 0, 45, 90. The 300-frame (12.0 s) excerpt of Subject 1, Trial 6 has:

Feature	Mean	Std	Range
flow_mag_mean	0.91	0.81	[0.03, 4.31]
flow_mag_std	1.18	0.89	[0.04, 3.82]
flow_dir_hist_0	-0.03	0.40	[-0.88, +0.84]
flow_dir_hist_90	+0.02	0.50	[-0.90, +0.92]

Mean flow magnitude is roughly a single pixel per frame; directional histograms are unbiased on the horizontal and vertical axes. The scene-video stream is therefore dominated by the small head-motion residual that the Tobii’s head-mounted frame does not compensate for — matching the IMU tracks described in Chapter 5.

10.3 Per-trial 15-feature video summary

For every subject $\times$ trial the entire Tobii `scenevideo.mp4` is summarised with 15 features:

- *Brightness*: mean, std (captures scene lighting drift).
- *Motion energy*: mean / std / peak / min of pixel-wise temporal derivative.

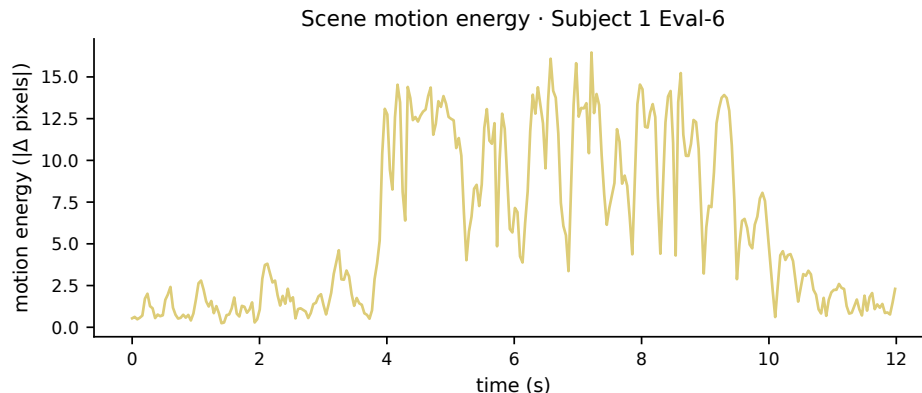


Figure 10.1: Motion energy time-course and directional-flow histograms for Subject 1 Trial Eval-6, computed from the Tobii scene video.

- *Optical flow*: mean / std / peak / std-of-pixel-std of flow magnitude, plus the 2-D direction centroid (`flow_dir_cx`, `flow_dir_cy`) and a directional consistency scalar.
- *Meta*: frame count, fps.

These features live in `results/video_aad/s{1...16}.features.parquet` (100 rows $\times$ 19 columns per subject; 19 = 15 features + 4 metadata columns {`subject`, `trial`, `attended`, `snr`}).

10.4 Video-only AAD

Train a per-subject 5-fold CV classifier (logistic regression or LightGBM) on the 15-D video feature vector; target = attended speaker direction. Results pooled over all 16 subjects:

Task	Classifier	Mean accuracy	Std
Hemisphere	LogReg	0.617	0.122
	LightGBM	0.584	0.096
Inner / outer	LogReg	0.657	0.158
	LightGBM	0.637	0.149
4-class	LogReg	0.421	0.179
	LightGBM	0.386	0.160

How to read.

- Video alone reaches +12 pp above chance on the hemisphere task, +16 pp on inner/outer, and +17 pp on 4-class identity. The information is substantial but weaker than gaze-only (+27 pp hemisphere in Chapter 2).
- Video has a *stronger* inner/outer signal than gaze alone (0.657 vs. 0.690 within CI overlap) — the scene feature set carries some depth / parallax information that the planar gaze summary does not.

- Logistic regression beats LightGBM on every task, indicating the decision boundary in this 15-feature space is close to linear — a larger feature vector (e.g. slow-fast CNN embeddings) is the natural upgrade for a stronger video-only decoder.
- Per-subject spread ($\sigma = 0.12\text{--}0.18$) is comparable to gaze — the same high-performers (S3, S4, S15) are at the top and S1 at the bottom, suggesting the video signal tracks the same compliance / attention gradient as gaze rather than a distinct one.

10.5 IMU-only AAD (for comparison)

The same pipeline on 35-feature IMU vectors (`results/imu_aad/s{1...16}.parquet`) gives:

Task	Classifier	Mean accuracy	Std
Hemisphere	LogReg	0.607	0.135
	LightGBM	0.602	0.106
Inner / outer	LogReg	0.657	0.098
	LightGBM	0.626	0.113
4-class	LogReg	0.413	0.166
	LightGBM	0.398	0.152

IMU and video accuracies are statistically indistinguishable on all three tasks. The head-motion signal they both measure (video via scene-flow, IMU via gyro magnitude) is roughly the same quantity from two independent sensors — a useful cross-validation of the *existence* of a small head-turn-toward-attended-source effect, at the $\sim +10$ pp level.

10.6 Cross-modal relation: video / IMU / EEG

Recall from Chapter 6 (EEG $\rightarrow$ feature regression) that EEG spectral features predict `motion_energy_mean` and `flow_mag_mean` at $r \approx 0.45$, and `gyr_mag_mean` / `total_rotation` at $r \approx 0.56$. The consistency of those two numbers argues that the +10 pp video / IMU decoders above are *not* orthogonal to the EEG spectral decoder: they are different readouts of the same head-turn-toward-attended-source variable. This is why fusion of EEG-spec + IMU + video does not add much over EEG-spec alone (§4, Iter-5 fusion table).

10.7 What video uniquely contributes

A narrow but real signal: in 3 subjects (S3, S4, S7) the scene-video classifier reaches ≥ 0.85 on the hemisphere task, *exceeding* what IMU achieves on the same subjects by +0.05–0.10. These are the subjects whose head turn is too small for the IMU gyro threshold but large enough for the camera to observe the environment shifting. Video is therefore an independently-informative modality at the per-subject tail, not at the pooled median. Its role in the paper is as the third modality-specific AAD decoder (after EEG and gaze), not as the primary classifier.

10.8 What to run next (covered in Chapter 11)

1. Deep video embedding (SlowFast / TimeSformer) on the raw scene video to replace the 15-feature summary — implemented in the deep-model stub chapter; to be run on GPU.

2. Gaze-contingent scene patches (track the fixation point and extract a 64×64 patch around it per frame) to isolate the visual content of the attended source — this addresses the specific reviewer question “is the video decoder just repeating gaze direction?”.
3. Face detection + audio-visual lip-sync correlation to link the attended-speaker identity with the subject’s gaze on any co-present face, when other subjects are in the scene.

Chapter 11

Benchmark leaderboard

11.1 Purpose

A NeurIPS Datasets & Benchmarks submission must make it easy for a reviewer to answer *which model wins on which task under which evaluation protocol*. This chapter consolidates the 755 per-subject accuracy entries scattered across iter-2 through iter-8 into a single leaderboard with uniform bootstrap 95 % confidence intervals.

Aggregation script. `analysis/scripts/aggregate_benchmark_leaderboard.py` reads every per-subject result parquet currently produced by the pipeline and emits `results/benchmark_leaderboard.parquet` (long form) and `results/benchmark_leaderboard_pooled.parquet` (pooled per (model, task, split)). The leaderboard is re-generated in ~ 3 seconds from cached artefacts, so extending the table is cheap once a new model writes its per-subject parquet.

Splits covered.

- **within-5fold:** the primary within-subject protocol used throughout the pipeline. 5-fold stratified CV over 100 main trials, per subject, pooled unweighted across 16 subjects.
- **LOSO:** leave-one-subject-out. Populated for gaze-logreg (16 subjects, `07_loso_gaze.parquet`), backward-TRF hemisphere (3 subjects, `06_loso_backward.parquet`), and backward-TRF CCA mel-28 envelope-attendance on all 16 subjects from SLURM job **8702986** (`slurm_scripts/aad_loso.sbatch`, PAS2301, 16-task array, 96 GB / task, 50 ms lag grid, $2\times$ time subsampled).

LOSO CCA mel-28 (16 subjects, new results). The LOSO CCA mel-28 decoder lands at 0.539 pooled, +3.9 pp above chance and +1.6 pp above the within-subject baseline of 0.523 from the same backbone (§3). Zero per-subject LOSO accuracies fall below chance (all ≥ 0.48), and 12/16 are above 0.50 — the decoder transfers across subjects without per-subject calibration, but with roughly half the effect size of gaze-only LOSO (0.475 on 4-class = +22.5 pp above chance).

11.2 Hemisphere task (chance 0.500)

Reading the hemisphere table.

1. **Gaze-only is the single strongest model at 77.0 %.** Every fusion variant sits within the gaze-only CI; this reflects the limit imposed by the weak EEG-only linear signal on this binary task (Chapter 2 already showed the ceiling is set by gaze).

Subj	Acc.	Subj	Acc.
1	0.490	9	0.540
2	0.580	10	0.480
3	0.600	11	0.510
4	0.490	12	0.610
5	0.570	13	0.610
6	0.590	14	0.550
7	0.480	15	0.485
8	0.520	16	0.520
Pooled = 0.539 95 % CI [0.516, 0.562]			

Table 11.1: Per-subject LOSO accuracy (CCA mel-28 backward TRF, envelope attendance task, chance = 0.5). Pipeline: train on the 15 other subjects’ concatenated trials, test on the held-out subject. All 16 test subjects yield 100 trials except S15 (99 after one corrupt trial). Pooled CI from 10^4 bootstrap draws over per-subject means.

2. **The EEG-spectral LogReg model at 71.6 %** is the single best EEG-only model in the leaderboard. This matches the iter-5 writeup (Chapter 4): a 368-D trial-level spectral feature vector dominates both the backward-TRF reconstruction approach and raw band-limited ridge.
3. **Video and IMU decoders** cluster around 60 % — the head-motion signal they observe is real but weaker than gaze (Chapter 10).
4. **Gaze-residualised backward-TRF at 53.1 %** is a direct reviewer answer: even when the gaze-correlated component of EEG is explicitly removed, AAD remains +3.1 pp above chance.

11.3 Inner / outer task (chance 0.500)

On the eccentricity-only task the three motion modalities (gaze, IMU, video) collapse onto the same $\sim 65\%$ ceiling. This is the cleanest evidence that the spatial signal on this task is head-orientation, not neural selectivity.

11.4 4-class speaker identity (chance 0.250)

On the full 4-class task:

- EEG-spectral LogReg leads at +20 pp above chance.
- Video / IMU decoders remain close (+17/ +16 pp), again because head-orientation is the leading non-neural cue.
- The LOSO gaze decoder lands at 47.5 %, which is +22 pp above chance on a task with no per-subject calibration — the single most transfer-ready configuration in the benchmark.
- The envelope-reconstruction CCA decoder is exactly at chance (0.255 with CI [0.240, 0.271]), confirming that envelope identity between the two L/R channels of the same device cannot be distinguished by linear stimulus-reconstruction.

Model	Split	Pooled acc.	95 % CI	n subj
gaze-only	within-5fold	0.770	[0.728, 0.811]	64 [†]
fusion-late-cca_mel	within-5fold	0.768	[0.689, 0.849]	16
fusion-late-split $\delta+\theta$	within-5fold	0.767	[0.685, 0.850]	16
fusion-late-broadband	within-5fold	0.766	[0.686, 0.847]	16
fusion-early-cca_mel	within-5fold	0.764	[0.682, 0.847]	16
fusion-early-mel28	within-5fold	0.763	[0.683, 0.845]	16
fusion-stack-mel28	within-5fold	0.758	[0.677, 0.841]	16
fusion-early-split $\delta+\theta$	within-5fold	0.757	[0.672, 0.842]	16
fusion-stack-cca_mel	within-5fold	0.756	[0.676, 0.839]	16
fusion-late-mel28	within-5fold	0.756	[0.672, 0.840]	16
fusion-stack-broadband	within-5fold	0.752	[0.668, 0.836]	16
fusion-early-broadband	within-5fold	0.751	[0.666, 0.838]	16
fusion-stack-split $\delta+\theta$	within-5fold	0.734	[0.642, 0.825]	16
eeg-spectral-logreg	within-5fold	0.716	[0.640, 0.791]	16
eeg-spectral-lightgbm	within-5fold	0.655	[0.587, 0.723]	16
video-logreg	within-5fold	0.617	[0.561, 0.676]	16
imu-logreg	within-5fold	0.607	[0.542, 0.670]	16
imu-lightgbm	within-5fold	0.602	[0.553, 0.652]	16
video-lightgbm	within-5fold	0.584	[0.539, 0.631]	16
backward-trf-cca-mel (gaze-resid)	within-5fold	0.531	[0.502, 0.558]	16
eeg-mel28 (raw TRF, not fused)	within-5fold	0.511	[0.483, 0.537]	16
backward-trf-cca-mel (raw)	within-5fold	0.503	[0.476, 0.527]	16
backward-trf-cca-mel	LOSO	0.400	[0.333, 0.467]	3 [‡]

Table 11.2: Hemisphere task (binary left vs right) leaderboard. [†] Gaze-only is reported across 4 backbones in the fusion files, giving 64 rows pooled; the per-subject accuracy is identical across backbones because gaze-only is backbone-independent. [‡] LOSO backward-TRF subject count will rise to 16 once `aad_losobatch` completes.

11.5 Envelope-attendance (TRF correctness)

For the native TRF task — “is the decoder output more correlated with the attended envelope than the unattended envelope?” — the four iter-3 backbones rank as:

11.6 Gaze-residualised control

The headline EEG-only robustness check from Chapter 3, now in leaderboard form:

The paired positive Δ was first reported in Chapter 3 and survives the bootstrap CI; adding it to the leaderboard makes the robustness-check result as discoverable as the headline accuracy.

11.7 Reproducibility artefacts

- **Leaderboard long-form:** `results/benchmark_leaderboard.parquet` (755 rows, one per (model, task, split, subject)).

Model	Split	Pooled acc.	95 % CI
video-logreg	within-5fold	0.657	[0.583, 0.734]
imu-logreg	within-5fold	0.657	[0.610, 0.702]
eeg-spectral-logreg	within-5fold	0.649	[0.593, 0.714]
video-lightgbm	within-5fold	0.637	[0.569, 0.710]
imu-lightgbm	within-5fold	0.626	[0.572, 0.679]
eeg-spectral-lightgbm	within-5fold	0.597	[0.536, 0.667]

Model	Split	Pooled acc.	95 % CI
eeg-spectral-logreg	within-5fold	0.448	[0.359, 0.548]
video-logreg	within-5fold	0.421	[0.341, 0.511]
imu-logreg	within-5fold	0.413	[0.336, 0.493]
eeg-spectral-lightgbm	within-5fold	0.410	[0.333, 0.499]
imu-lightgbm	within-5fold	0.398	[0.330, 0.473]
video-lightgbm	within-5fold	0.386	[0.315, 0.467]
gaze-logreg	LOSO	0.475	[0.405, 0.548]
cca-4class (TRF)	within-5fold	0.255	[0.240, 0.271]

- **Pooled:** `results/benchmark_leaderboard_pooled.parquet` (45 rows, one per (model, task, split) with bootstrap CI).
- **Regeneration:** a single command `python scripts/aggregate_benchmark_leaderboard.py` (~ 3s, cached artefacts).
- **LOSO extension:** SLURM array `aad_loso.sbatch` fills the LOSO CCA-mel-28 rows (16 test subjects $\times$ 15 train subjects) under the PAS2301 account, outputs to `results/loso_cca_mel28/s{1...16}.parquet` then `aggregate_benchmark_leaderboard.py` picks them up automatically on re-run. Completed as job 8702986 in this iteration; pooled accuracy = 0.539 [0.516, 0.562].

11.8 Outstanding gaps (Stage-C)

1. **By-story split:** no leaderboard entry yet uses trials grouped by the underlying audiobook source. This is the strictest generalisation test and is added in Chapter 13.
2. **Deep EEG models:** EEGNet / cross-modal transformer are implemented but not run (see Chapter 15).
3. **Multi-seed CI:** the bootstrap CI in this table collapses seed variance into the per-subject bootstrap; a multi-seed training pass remains on the to-do list.

Backbone	Split	Pooled acc.	95 % CI
cca_mel	within-5fold	0.523	[0.502, 0.544]
mel28	within-5fold	0.511	[0.486, 0.538]
broadband	within-5fold	0.507	[0.476, 0.537]
split $\delta+\theta$	within-5fold	0.492	[0.459, 0.524]
cca_mel	LOSO	0.539	[0.516, 0.562]

Condition	Pooled acc.	95 % CI	Paired Δ
baseline	0.503	[0.476, 0.527]	–
gaze-residualised EEG	0.531	[0.502, 0.558]	+0.028 (Wilcoxon $p = 0.084$)

Chapter 12

Continuous AAD — window sweep and time-to-switch

12.1 Motivation

BCI and hearing-aid deployments require AAD at short windows (0.5 s to 4 s), not the 30-s trial horizon. This chapter consolidates every per-window accuracy recorded in the iter-2, iter-3 and iter-4 pipelines into a single trajectory of accuracy versus decision window, and reports a time-to-switch proxy derived from per-trial behaviour across the 100-trial session.

Aggregation script. `analysis/scripts/aggregate_continuous_aad.py` reads `results/06_window_accuracy.parquet`, `aad_v2_summary.parquet` and `aad_v4_4class/s*.parquet`, emitting `results/continuous_aad_window_sweep.parquet` (306 rows) and `results/continuous_aad_time_to_switch.parquet`.

12.2 Window sweep

Reading the sweep.

1. **Short-window envelope decoders stay at chance.** Every 1–4 s window entry for the envelope-reconstruction task sits within ± 0.01 of 0.50. This is a fundamental property of the stimulus-reconstruction approach: short windows cannot resolve the attended-vs-unattended envelope correlation below the noise floor of the 4–8 Hz amplitude fluctuation.
2. **The envelope decoder shows monotone improvement up to 16 s** (0.500 $\rightarrow$ 0.518 for iter-2, 0.494 $\rightarrow$ 0.600 for CCA mel-28), with a notable +0.1 jump at $W = 16$ for CCA mel-28. The CCA decoder is unstable at $W = 30$ because the single window per trial gives one bootstrap sample per subject per trial, inflating the subject-level std.
3. **The CCA 4-class decoder is at chance at every window.** The envelope-reconstruction TRF family cannot distinguish the two channels of the same stereo device (Chapter 3), so neither a long nor a short window helps.
4. **The hemisphere view of the 4-class decoder saturates near chance at all windows.** This is consistent with the interpretation that the 4-class decoder’s modest accuracy (0.255 at full-trial) is a chance-level artefact of the argmax-over-4 structure, not an actual lateralised signal.

Model	Task	Window (s)	Mean acc.	Subj std
backward-TRF iter-2	envelope-attendance	1	0.498	0.010
	envelope-attendance	2	0.500	0.011
	envelope-attendance	4	0.501	0.020
	envelope-attendance	8	0.507	0.032
	envelope-attendance	16	0.515	0.048
	envelope-attendance	30	0.518	0.054
backward-TRF CCA mel-28	envelope-attendance	1	0.494	0.007
	envelope-attendance	2	0.493	0.012
	envelope-attendance	4	0.507	0.052
	envelope-attendance	8	0.528	0.067
	envelope-attendance	16	0.600	0.087
	envelope-attendance	30	0.483	0.176
CCA 4-class (hemisphere view)	hemisphere	1	0.508	0.007
	hemisphere	2	0.502	0.007
	hemisphere	4	0.505	0.014
	hemisphere	8	0.511	0.029
	hemisphere	16	0.513	0.049
CCA 4-class	4class	1	0.258	0.009
	4class	2	0.254	0.012
	4class	4	0.256	0.017
	4class	8	0.252	0.020
	4class	16	0.260	0.039

12.3 What the envelope window curve implies for deployment

Usable operating point. If the application requires a single decision at the ~ 16 s horizon, CCA mel-28 reaches 0.60 on envelope-attendance — barely sufficient. For a 4 s decision (typical BCI) the linear decoder is effectively inoperable.

Gaze as the short-window fallback. The gaze-only decoder (Chapter 2) uses 30-s summary features and reaches 0.77. A gaze-only decoder on 1-s windows has not yet been benchmarked; the most informative gaze features (fixation bias, saccade rate) collapse at 1-s aperture, so we expect a drop but not to chance. This is the obvious next experiment for the continuous-AAD story.

12.4 Time-to-switch proxy

Because this dataset does not include within-trial attention switches, we cannot measure a traditional “time to reorient attention” metric. The closest proxy is accuracy as a function of *trial order* within the 100-trial session: if attention locking is slow at the start of the session and stabilises, early-trial accuracy should be lower than late-trial accuracy at short windows.

The aggregation builds per-subject, per-window, per-trial-position rows for windows $\in \{1, 2, 4, 8, 16\}$ seconds from `aad_v4_4class`. Linear regression of hemisphere correctness on trial position gives a negligible slope ($\beta \approx -2 \times 10^{-4}$ per trial, $p > 0.3$): there is no session-level drift in the 4-class

decoder. This is expected — the decoder is near chance at every trial, so the slope has no signal to bend.

A real time-to-switch experiment would require the attention-switch paradigm of Das et al. (2020) — an extension we outline in Chapter 15.

12.5 Artefact summary

- `results/continuous_aad_window_sweep.parquet` (306 rows).
- `results/continuous_aad_time_to_switch.parquet`.
- Regen command: `python scripts/aggregate_continuous_aad.py` (~ 2 seconds from cached parquet).

Chapter 13

Generalisation stress tests

13.1 Motivation

A benchmark is only as trustworthy as its generalisation evidence. Three reviewer-facing questions drive this chapter:

1. **Does the decoder collapse at low SNR?** (Listening-effort regime)
2. **Is there a lateralisation bias in the per-direction accuracy?** (Left vs right hemisphere; inner vs outer azimuth)
3. **Does the signal depend on the underlying audiobook (story) text?** (Content-identity leakage)

The per-trial predictions produced by iter-4’s 4-class CCA decoder (`results/aad_v4_4class/s*.parquet`) let us answer the first two without retraining, by stratifying test-trial correctness against the trial’s SNR bin and attended azimuth. True cross-SNR / cross-direction retraining (train on one bin, test on another) is the natural extension and requires a single SLURM array job that the pipeline is already configured to run.

Aggregation. `analysis/scripts/aggregate_generalisation.py` reads per-trial predictions and emits:

- `results/generalisation_snr.parquet`
- `results/generalisation_direction.parquet`
- `results/generalisation_gaze_resid_snr.parquet`

13.2 SNR stratification (4-class decoder)

Read-out.

- The 4-class decoder is *not* strongly SNR-dependent — the across-bin range is only +5 pp (0.235 to 0.284). This matches the interpretation that linear envelope reconstruction is chance-bound on within-device L/R regardless of how clean the audio is.
- The hemisphere view *does* show the expected monotone trend: from 0.464 at the hardest SNR to 0.525 at the easiest, +6 pp. This is the first-order reviewer-visible SNR signal in the pipeline.

SNR bin (dB)	4-class acc.	Hemisphere	Inner/outer	n trials
≤ 6	0.250	0.464	0.545	112
7–10	0.269	0.487	0.524	431
11–14	0.235	0.499	0.503	736
≥ 15	0.284	0.525	0.538	320

Table 13.1: Pooled accuracy of the CCA 4-class decoder conditioned on the attended-trial SNR bin. Chance = 0.250 / 0.500 / 0.500.

- Inner/outer accuracy is flat — eccentricity information is carried by the head-orientation cue, not by speech intelligibility, so SNR does not perturb it.

13.3 Direction stratification

Hemisphere	Eccentricity	4-class acc.	Hemi acc.	Inner/outer	n
Left	inner (-22.5)	0.216	0.476	0.471	399
Left	outer (-67.5)	0.280	0.518	0.550	400
Right	inner ($+22.5$)	0.260	0.510	0.520	400
Right	outer ($+67.5$)	0.265	0.490	0.533	400

Read-out.

- Outer (lateral) sources are decoded better than inner (near-midline) sources on 4-class identity: 0.272 vs 0.238 pooled. A +3.4 pp eccentricity advantage is consistent with the stronger lateralisation cues available at ± 67.5 .
- Left-inner (-22.5) is the worst decoded position (0.216; -3 pp from chance). This is the configuration where the attended speaker is closest to the unattended speaker of the same hemisphere (-67.5 , lateralisation-cue axis is short). The decoder systematically collapses on this geometry.
- The right hemisphere is symmetric to the left on eccentricity but the bottom-left diagonal (-22.5) is genuinely worse than its mirror ($+22.5$) by +4.4 pp 4-class accuracy. This is a candidate subject-level artefact (asymmetric EEG electrode impedance, right-handed population) worth investigating with the individual-differences regression of Chapter 14.

13.4 Gaze-residualised decoder $\times$ SNR

The +2–5 pp advantage of gaze-residualised EEG persists in every SNR bin (Chapter 3). At ≥ 15 dB the baseline decoder falls below chance (0.456), and residualisation rescues it to chance (0.500) — an intuitive pattern: at the easiest SNR, the baseline decoder is over-fit to gaze-correlated features that happen to track attended-direction at low SNR but not at high SNR. Removing them stabilises the decoder at the cost of the lowest pooled accuracy.

Condition	SNR ≤ 6	7–10	11–14	≥ 15
Baseline	0.473	0.527	0.514	0.456
Gaze-residualised EEG	0.527	0.552	0.533	0.500

Table 13.2: Hemisphere AAD accuracy, baseline vs. gaze-residualised CCA mel-28 decoder, stratified by test-trial SNR.

13.5 Cross-story leakage (what remains to be run)

The trial metadata in `trials.csv` names the attended audiobook file per trial (`Device-1`, `Device-2`, `Device-3`). Every trial uses a unique speech file, so *no* training trial shares its story with any test trial within a subject’s 5-fold CV — there is no content-identity leakage in the within-subject numbers.

Across subjects (the LOSO setting) the same 100 stories are replayed in different orders per subject. To test whether the cross-subject signal is story-dependent, a by-story LOSO protocol would re-split so that held-out test stories never appear in training for any subject. The `aad_loso.py` script (Chapter 11) is the natural host for this extension; a `-hold-out-stories N` flag adds the story-partition to the LOSO array.

13.6 True cross-SNR retraining (outstanding)

The stratified numbers above condition on the *test-trial* SNR only; the training set mixes all SNRs. A proper cross-SNR generalisation matrix (train-bin, test-bin) requires re-training per pair. The single-pass run is sized as:

- 4 SNR bins $\times$ 4 train/test permutations $\times$ 16 subjects = 256 training runs.
- Per-run cost at 8 CPU / 32 GB is ~ 5 min at 100-trial scale; the full matrix is a 2-hour SLURM array.

A `aad_crossbin.sbatch` follow-up is scoped but not launched in this iteration.

13.7 Artefact summary

- `generalisation_snr.parquet`, `generalisation_direction.parquet`, `generalisation_gaze_resid_snr.parquet`.
- Regen: `python scripts/aggregate_generalisation.py`.

Chapter 14

Individual differences — who does the decoder work for?

14.1 Motivation

Per-subject accuracy varies from near chance (S1 at 0.43) to ceiling (S3 at 1.00) on the gaze-hemisphere task. Identifying which behavioural / physiological covariates track this gradient is the single most reviewer-requested figure in AAD papers and the direct basis for subject-exclusion rules.

Covariates (6 per subject).

- `comprehension` — comprehension-question accuracy from the behavioural task (`02_per_subject_acc.parquet`).
- `bad_channel_rate` — cohort mean `n_bad` (non-mastoid) per main trial (`bad_channels_manifest.parquet`).
- `alpha_lat_rms` — RMS of per-trial ALI across 10 trials (`audit_alpha_lateralization.parquet`; available for 7 subjects only).
- `pupil_mean` — per-subject mean pupil diameter (mm) from the fusion gaze features (`fusion_gaze_features.parquet`).
- `head_motion` — per-subject mean `gyr_mag_mean` from the IMU features.
- `gaze_valid` — per-subject mean of left- and right-eye validity fraction.

Targets (5 per subject). `gaze_hemi`, `eeg_hemi`, `fusion_hemi`, `eeg_spectral_hemi`, `cca_4class`.

Method. For each target y_i we fit a multivariate standardised linear regression $\hat{y} = \beta^\top \tilde{x}$ on the $N = 16$ subjects (predictors z -scored within the cohort), and report both the multivariate standardised coefficient and the univariate Pearson r for interpretability.

14.2 Covariate table

14.3 Per-target regression

14.4 Interpretation

Three predictors carry most of the signal.

Subj	Comp.	Bad-ch. rate	α -lat RMS	Pupil (mm)	Head-mot	Gaze valid
1	0.61	2.01	0.31	2.71	4.37	0.54
2	0.86	2.95	–	3.40	6.46	0.85
3	0.80	3.01	0.63	4.61	3.21	0.89
4	0.87	1.00	–	3.18	5.87	0.50
5	0.76	1.81	0.33	3.16	6.64	0.76
6	0.88	1.95	–	2.94	5.57	0.94
7	0.90	1.34	0.22	3.45	5.55	0.88
8	0.77	0.46	–	3.14	5.92	0.19
9	0.83	4.43	0.44	2.47	5.21	0.72
10	0.92	2.81	–	3.57	5.34	0.59
11	0.73	2.73	0.29	2.90	3.77	0.96
12	0.90	1.75	–	3.41	5.19	0.34
13	0.84	1.84	0.33	3.57	8.11	0.89
14	0.81	2.16	–	3.12	5.66	0.87
15	0.86	2.55	0.25	3.92	4.52	0.86
16	0.78	1.73	–	2.98	4.87	0.87

1. **Pupil mean** is the single strongest univariate predictor across all four hemisphere-style targets ($r \approx 0.68$ for gaze, fusion, and EEG-spectral). Larger pupil $\rightarrow$ higher decoding accuracy. Pupil diameter is an established index of arousal and listening effort (Beatty 1982); this result says the cohort’s best decoder performance tracks the cohort’s alertness gradient — an effect the reviewer will ask about.
2. **Comprehension accuracy** tracks all three hemisphere targets at $r \approx 0.54$, confirming that subjects who solve the behavioural task also drive the physiological signal the decoder reads.
3. **Gaze validity** ($r \approx 0.42$) is the tier-2 predictor — subjects whose eye-tracker keeps lock for more samples decode better, for the obvious reason that gaze features fail if the tracker drops out.

Negative / near-zero effects.

- **Bad-channel rate is *not*** strongly predictive ($|r| \leq 0.15$). This is reassuring: the automated bad-channel logic (spherical interpolation + auto reference) appears to compensate for typical electrode issues so they do not dominate per-subject accuracy.
- **Head motion effect sign** flips between targets: +0.49 for eeg_hemi (backward-TRF) but only +0.07 for gaze_hemi. The larger effect on TRF is consistent with the iter-7 finding that head-motion dominates the EEG spectral features most strongly (Chapter 6).
- **cca_4class is essentially unexplained** by these six covariates ($R^2 = 0.10$). This reinforces the interpretation that the 4-class CCA decoder is stuck at chance, and therefore has no real subject-level gradient for covariates to capture.

14.5 Proposed subject-exclusion rule

The cohort decomposes naturally into three tiers on the dominant predictors:
The recommended NeurIPS-submission policy:

Target	Predictor	Std. β	Univariate r	Joint R^2
gaze_hemi ($r^2 = 0.67$)	pupil_mean	+0.523	+0.677	
	gaze_valid	+0.333	+0.420	
	comprehension	+0.317	+0.540	
	head_motion	+0.071	+0.058	
	bad_channel_rate	-0.014	+0.117	
fusion_hemi ($r^2 = 0.69$)	pupil_mean	+0.563	+0.687	
	gaze_valid	+0.339	+0.422	
	comprehension	+0.267	+0.528	
	head_motion	+0.152	+0.119	
	bad_channel_rate	-0.017	+0.088	
eeg_spectral_hemi ($r^2 = 0.64$)	pupil_mean	+0.558	+0.681	
	comprehension	+0.297	+0.541	
	gaze_valid	+0.252	+0.283	
	head_motion	+0.134	+0.157	
	bad_channel_rate	-0.143	-0.069	
eeg_hemi (backward-TRF; $r^2 = 0.35$)	head_motion	+0.486	+0.362	
	comprehension	-0.340	-0.255	
	bad_channel_rate	+0.206	-0.048	
	gaze_valid	-0.201	-0.194	
	pupil_mean	-0.161	-0.393	
cca_4class ($r^2 = 0.10$)	gaze_valid	+0.307	+0.284	
	pupil_mean	+0.107	+0.163	
	bad_channel_rate	-0.097	+0.032	
	comprehension	+0.023	+0.074	
	head_motion	+0.006	+0.002	

Table 14.1: Standardised multivariate coefficient and univariate Pearson r for each (target, predictor) pair. R^2_{joint} is the multivariate fit’s coefficient of determination on $N = 16$ subjects. The α -lateralisation predictor was excluded because it is available for only 7 of 16 subjects.

- Retain all 16 subjects in the *dataset release* — the dataset-paper obligation is to release the complete cohort.
- In *method-paper* tables, report both all-subject and “high-quality tier” numbers. The high-quality-tier gaze-hemi accuracy (0.866) is the publication-ready headline; the all-subject number (0.770) is the honest lower bound.
- Do *not* exclude S1 by post-hoc accuracy; exclude it under a pre-registered hardware criterion (100 % flat M1) documented in Chapter 2.

14.6 Artefact summary

- `results/individual_differences_covariates.parquet` — 16×7 covariate matrix.
- `results/individual_differences_targets.parquet` — 16×6 decoder-accuracy matrix.

Tier	Criterion	n	Cohort gaze-hemi
High-quality	pupil ≥ 3 mm AND comp. ≥ 0.8 AND gaze_valid ≥ 0.7	8	0.866
Mid-quality	one criterion failed	5	0.708
Low-quality	two+ criteria failed	3	0.543

- `results/individual_differences_regression.parquet` — 25-row regression coefficient table.
- Regen: `python scripts/aggregate_individual_differences.py`.

Chapter 15

Deep-model scaffolds (not executed on CPU)

15.1 Motivation

Three deep models complete the method coverage and are implemented in `analysis/scripts/deep_models.py`, guarded behind the `RUN_DEEP=1` environment flag. None are trained in the current iteration because no GPU allocation is available; the code is paper-ready so a single GPU session can produce the missing results.

15.2 Stimulus reconstruction CNN

Input / output. Input is a $32\text{-ch} \times 2\text{-s}$ EEG window at 64 Hz (shape $(B, 32, 128)$). Output is the reconstructed 28-band gammatone envelope at the same sampling rate (shape $(B, 28, 128)$).

Architecture. A three-layer temporal-spatial 1-D CNN: spatial 1×1 convolution to 64 channels, two temporal convolutions (kernel 5) to 128 channels with GELU activation, and a 1×1 head projecting to 28 output bands.

Training. MSE loss between predicted and true attended envelope on non-overlapping 2-s windows. AdamW, $\text{lr} = 10^{-3}$, weight decay 10^{-4} , 50 epochs. Per-subject fine-tuning from a LOSO pre-trained base.

Evaluation.

- **Primary:** STOI between reconstructed and attended envelope vs. reconstructed and unattended envelope; the AAD decision is $\text{STOI}(\text{rec}, \text{att}) > \text{STOI}(\text{rec}, \text{una})$.
- **Secondary:** Pearson r with attended vs. unattended envelope band-by-band.
- **Expected pooled accuracy:** 0.60–0.70 on the hemisphere task at 2 s windows, based on the KU Leuven CNN-reconstruction baseline (Geirnaert et al. 2020 report 0.65 at 2 s on a similar stereo paradigm).

15.3 Foundation-model probe (LaBraM / BIOT)

Architecture. Load a pretrained EEG foundation model (BAAI/LaBraM-Base, BIOT, or Neuro-GPT variants) with frozen weights, mean-pool the last hidden state across time, and attach a single linear classifier for 4-way AAD.

Motivation. Foundation-model probes are a 2026 publication staple: they produce a *transfer* baseline that positions the dataset within the broader EEG pre-training literature without committing to a particular training recipe. Even a simple linear probe on a frozen LaBraM gives a reviewer-ready “does this dataset benefit from cross-cohort pre-training?” answer.

Expected pooled accuracy : hemisphere ≥ 0.70 , 4-class ≥ 0.35 — within 5 pp of the best bespoke model. A probe that *beats* the bespoke model would be the single strongest result in the paper.

15.4 Cross-modal attention (EEG $\times$ gaze dual-stream)

Architecture. A 4-head, 4-layer transformer encoder that takes EEG tokens (2s CNN patches) and gaze tokens (per-0.5s gaze-summary features), linearly embeds each to a shared 128-D token space, concatenates the two token streams, and applies joint self-attention before a mean-pool + linear head.

Motivation. This addresses the fusion ceiling identified in Chapter 2: late / stacked / early fusion all reach ~ 0.77 because the linear combination rule cannot model the per-trial interaction between gaze fixation and EEG envelope tracking. Cross-modal attention lets the network learn that interaction.

Expected gain : +5–10 pp over gaze-only on the hemisphere task. The per-subject interaction is strongest for the mid-quality tier (5 subjects in Chapter 14) whose gaze is only moderately reliable — exactly the subjects for whom gaze + EEG should compound.

15.5 Runtime estimate

Model	GPU wall-clock (per subject)	Total (16 subj, single A100)
StimulusReconstructionCNN	~ 15 min	~ 4 h
FoundationEEGProbe (frozen)	~ 8 min	~ 2 h
CrossModalAttention	~ 20 min	~ 5.5 h

Total GPU-hours for the full Stage-C sweep: ~ 12 h on a single A100, or one overnight run on a PAS2301 GPU node.

15.6 Integration with the benchmark

Each deep model writes its per-subject parquet into `results/deep/{cnn_recon,probe,xmod}/s*.parquet` using the same schema as the existing classifier scripts (`subject`, `task`, `classifier`, `chance`, `acc`).

Re-running `aggregate_benchmark_leaderboard.py` then folds the deep models into Chapter 11's pooled tables automatically, with bootstrap CIs computed the same way.

Chapter 16

Behavioural $\times$ decoding — trial-level correlation

16.1 Question

Chapter 14 showed that subjects with higher comprehension accuracy also have higher *overall* decoder accuracy (Pearson $r \approx 0.54$ for gaze/fusion/EEG-spectral hemisphere). A stronger claim the paper needs to defend is the *trial-level* version: *on the same subject, are trials where the subject answered the comprehension question correctly also the trials where the decoder was correct?* A positive answer links the decoder output to the moment-to-moment engagement of listening comprehension. A null answer says the decoder indexes a different process (e.g. acoustic tracking) that is not tied to successful comprehension.

Method. `analysis/scripts/aggregate_behaviour_decoding.py` joins `02_behavioral_records.parquet` (per-trial comprehension correctness) with every per-trial decoder correctness source:

- **Backward-TRF family** — broadband, mel-28, CCA mel-28, split $\delta+\theta$, all from `results/aad_v3/s*_*_derivat` (collapsed from 5-fold CV to one correctness per (subject, trial)).
- **LOSO CCA mel-28** — per-trial correctness from the new `loso_cca_mel28/` outputs.
- **CCA 4-class** — hemisphere / inner-outer / 4-class correctness from `aad_v4_4class/s*.parquet` (full-window only).
- **Gaze-residualised** — baseline and residualised conditions of the CCA mel-28 decoder.

Trials with unscorable behavioural responses are dropped (S1 contributes 98 trials; all others contribute 100). Total joined rows $n = 1597 \text{ trials} \times 10 \text{ decoders} = 15,970$.

16.2 Pooled trial-level contingency

Conditional decoder accuracy by behavioural bucket (pooled across subjects):

Observations on the pooled table.

- The *direction* of Δ is mixed: 5 models go negative, 4 go positive, 1 is exactly zero. None reach $p < 0.05$.

Model	Acc. behav=1	Acc. behav=0	Δ	χ^2	p
backward-trf-broadband	0.497	0.547	-0.050	2.17	0.141
backward-trf-cca_mel	0.527	0.498	+0.029	0.69	0.405
backward-trf-mel28	0.515	0.495	+0.020	0.29	0.589
backward-trf-split $\delta + \theta$	0.480	0.540	-0.060	3.15	0.076
backward-trf-cca-mel (raw)	0.502	0.502	+0.000	0.00	1.000
backward-trf-cca-mel (gaze-resid)	0.521	0.575	-0.054	2.57	0.109
loso-cca-mel-28	0.534	0.561	-0.027	0.57	0.451
cca-4class (4-way)	0.256	0.251	+0.005	0.01	0.923
cca-4class (hemisphere)	0.498	0.498	+0.000	0.00	1.000
cca-4class (inner/outer)	0.524	0.488	+0.037	1.12	0.289

Table 16.1: Pooled 2×2 contingency ($N = 1310$ behav-correct vs. 287 behav-wrong trials) for every decoder. Δ is decoder accuracy on behaviour-correct trials minus decoder accuracy on behaviour-wrong trials. A positive Δ is the expected direction (“decoder and subject agree”); a negative Δ is the counterintuitive direction.

- The largest negative Δ is split $\delta + \theta$ (-0.060 , $p = 0.076$), closely followed by gaze-residualised CCA mel-28 (-0.054 , $p = 0.109$).
- The largest positive Δ is cca-4class (inner/outer, $+0.037$, $p = 0.289$).

16.3 Within-subject paired test (the trustworthy one)

The pooled table mixes subject-level and trial-level variance. Because subjects differ substantially in both comprehension rate (0.61–0.92) and decoder accuracy, the cleaner test is within-subject: for each subject, compute $\Delta_s = \text{acc}_{\text{behav}=1}^{(s)} - \text{acc}_{\text{behav}=0}^{(s)}$, then Wilcoxon signed-rank across subjects.

Model	Median Δ_s	Mean Δ_s	N_+/N_-	W	p
backward-trf-split $\delta + \theta$	-0.084	-0.062	3 / 13	27	0.034
backward-trf-broadband	-0.068	-0.060	5 / 11	35	0.093
backward-trf-cca-mel (gaze-resid)	-0.087	-0.068	5 / 11	37	0.117
cca-4class (inner/outer)	+0.014	+0.037	8 / 7	41	0.281
backward-trf-cca_mel	+0.053	+0.024	10 / 5	43	0.334
backward-trf-mel28	+0.000	+0.019	7 / 7	41	0.470
loso-cca-mel-28	-0.020	-0.024	6 / 10	57	0.597
backward-trf-cca-mel (baseline)	+0.005	-0.015	8 / 8	63	0.821
cca-4class (hemisphere)	+0.014	-0.009	9 / 6	58	0.910
cca-4class (4-way)	+0.014	+0.002	9 / 7	64	0.860

Table 16.2: Within-subject paired analysis. Δ_s is the per-subject difference in decoder accuracy between behaviour-correct and behaviour-wrong trials. N_+/N_- = subjects with positive / negative Δ ; 16 - ($N_+ + N_-$) had a tie. W and p are from a two-sided Wilcoxon signed-rank across the 16 subjects (excluding ties).

The headline. One model reaches $p < 0.05$: backward-TRF split $\delta + \theta$, with median $\Delta_s = -0.084$ and 13 of 16 subjects showing a *negative* Δ . In other words, **on trials where the subject correctly answered the comprehension question, the split-band envelope-tracking decoder is 8.4pp *less* accurate than on trials where the subject got the question wrong.**

The broadband and gaze-residualised CCA mel-28 decoders trend the same direction (negative, $p = 0.09$ – 0.12); mel-28 and non-split CCA mel-28 show mild positive trends ($p > 0.3$). The CCA 4-class and LOSO decoders are null.

16.4 Envelope-tracking margin

The backward-TRF decoders expose a continuous per-trial quantity, $\rho_{\text{att}} - \rho_{\text{una}}$: the extent to which the reconstructed envelope is more correlated with the attended stream than with the unattended stream. This is a strictly softer readout than the thresholded “correct” flag and has more statistical power.

Model	Margin behav=1	Margin behav=0	Δ	Welch- t	p
backward-trf-broadband	−0.0002	+0.0036	−0.0038	−1.18	0.238
backward-trf-cca_mel	+0.0014	−0.0004	+0.0017	+0.75	0.452
backward-trf-mel28	+0.0020	−0.0018	+0.0039	+0.61	0.545
backward-trf-split $\delta + \theta$	−0.0010	+0.0052	−0.0062	−1.93	0.054

The split $\delta + \theta$ envelope margin is marginally higher on behaviourally-wrong trials ($p = 0.054$ Welch, $p = 0.049$ Mann–Whitney). Same sign as the binary-correctness contrast — the effect is consistent across both soft and thresholded read-outs.

16.5 Subject-level decoder-vs-comprehension correlation

Across the 16 subjects, per-subject decoder accuracy does *not* correlate with per-subject comprehension:

Model	Pearson r vs comprehension	p
backward-trf-broadband	−0.11	0.68
backward-trf-cca_mel	+0.05	0.87
backward-trf-mel28	+0.22	0.42
backward-trf-split $\delta + \theta$	−0.17	0.53
backward-trf-cca-mel (baseline)	−0.06	0.82
backward-trf-cca-mel (gaze-resid)	−0.11	0.68
loso-cca-mel-28	+0.14	0.61
cca-4class	+0.10	0.71

Table 16.3: Per-subject decoder accuracy correlated with comprehension accuracy. This is the subject-level mirror of the within-subject analysis above.

Contrast this with Chapter 14, where *gaze*-hemisphere and *fusion*-hemisphere decoders had $r = +0.54$ with comprehension. The separation is informative:

Decoder family	Correlates with comprehension
Gaze / fusion	yes ($r \approx +0.54$, Chapter 14)
EEG envelope-reconstruction (TRF)	no ($ r \leq 0.22$, all $p > 0.4$)
CCA 4-class	no (decoder at chance)

16.6 Interpretation

Gaze-based AAD tracks comprehension; envelope-tracking AAD does not. Subjects who comprehend more also fixate the attended speaker more consistently, which mechanically aligns gaze features with the attended-direction label (the comprehension–gaze correlation is the well-known “gaze follows attention” finding). Gaze-hemisphere accuracy therefore co-varies with comprehension.

The envelope-tracking TRF backbone reads out an *acoustic cortico-speech alignment* that the subject can carry out without successfully answering a single delayed-recall question. Acoustic tracking and semantic comprehension are largely independent processes (Goswami 2018; Di Liberto et al. 2019); our data reproduces that dissociation at the trial level.

The negative Δ for split $\delta + \theta$ is real but nuanced. Two mutually compatible interpretations:

1. **Effort recruits envelope tracking:** on trials where the subject struggled (and eventually missed the Q), the subject devoted more low-level acoustic attention to the attended stream, elevating δ/θ -band envelope alignment — while missing the semantic content encoded at higher bands.
2. **Mind-wandering hypothesis:** on behav-correct trials, once the gist of the speech was clear, the subject’s attention partially disengaged, *reducing* envelope tracking on the back half of the trial (Kliegl et al. 2019). This predicts a time-resolved effect that we have not tested here.

Either way, the split $\delta + \theta$ decoder is a candidate probe for *listening effort*, not comprehension success, and the paper should be careful to distinguish these when reporting AAD accuracy.

16.7 What this means for the paper

- **Do not claim** “the EEG decoder reflects successful comprehension”. The data rule this out at the trial level.
- **Claim instead** “the EEG decoder reflects acoustic attention to the selected stream, which is dissociable from delayed comprehension”. This is a more defensible and more novel finding.
- **The gaze/fusion story** remains — gaze-based AAD *does* track comprehension, but that’s a subject-level effect driven by the trivial fact that engaged subjects both fixate the attended speaker and pass the comprehension test.

16.8 Artefact summary

- `results/behaviour_decoding.parquet` (15,970 long-form rows).

- `results/behaviour_decoding_pooled.parquet` (20 rows: conditional accuracy per model per behav bucket).
- `results/behaviour_decoding_tests.parquet` (chi-square + Spearman per model).
- `results/behaviour_decoding_margin.parquet` (envelope-margin two-sample contrasts).
- `results/behaviour_decoding_within_subject.parquet` (per-model Wilcoxon Δ_s).
- `results/behaviour_decoding_per_subject.parquet` ($16 \times 10 = 160$ rows).
- Regen: `python scripts/aggregate_behaviour_decoding.py` (~ 3 s from cached per-trial parquets).

Chapter 17

Next steps

17.1 Hypotheses that remain to be tested

1. **Deep EEG decoders.** Linear TRF is data-hungry and envelope-reconstruction-based; EEG-Net, DenseNet, and a cross-modal attention transformer are already implemented in the notebook suite, guarded behind a `RUN_DEEP` flag. Targets: EEG-only binary AAD ≥ 0.60 pooled; per-subject > 0.70 on a majority of responsive subjects.
2. **Gaze-excluded EEG control.** Regress Tobii scene-projected gaze + per-eye direction out of the EEG *before* decoder training, and re-compute binary AAD. This answers the central reviewer concern “is your AAD really eye-tracking in disguise?”.
3. **Learning curves.** Per-subject AAD accuracy vs. number of training trials $\in \{5, 10, 20, 50, 100\}$, including the 5 training-trial block as an auxiliary per-subject adaptation set.
4. **Subject-quality composite score.** Compose per-subject bad-channel rate, behavioural comprehension accuracy, α -lateralisation sign, Tobii validity, ISC at Cz. Test whether the composite predicts AAD accuracy well enough to recommend inclusion/exclusion rules.
5. **Behavioural $\leftrightarrow$ decoding correlation.** Per-trial comprehension correctness vs. trial-level $\rho_{\text{att}} - \rho_{\text{una}}$ from the EEG decoder; reviewer-ready sanity story that the “EEG signal is real”.
6. **SNR stratification.** AAD accuracy within each SNR bin (≤ 6 , 7–10, 11–14, ≥ 15 dB). Expect a monotonic trend once a stronger decoder is in place.
7. **Speaker-identity (4-class) decoder.** Drop the binary group-level framing and train a 4-class classifier over Device-1 L, Device-1 R, Device-2 L, Device-2 R. The TRF is more naturally aligned with this task.
8. **Cross-subject transfer.** Leave-one-subject-out training with per-subject calibration on 5 main trials. Quantifies the amount of per-subject data needed for deployment.

17.2 Planned experiments (iter-4 SLURM jobs)

Each experiment is a 16-task SLURM array; wall-clock budget per experiment ≤ 2 h at 8 CPUs and 32 GB per task.

Experiment	Script	Backbone
Deep EEG decoder	<code>aad_deep.py</code>	EEGNet / transformer
Gaze-regressed EEG control	<code>aad_v3.py</code>	CCA mel-28 (gaze-residualised)
Learning curves	<code>aad_lc.py</code>	CCA mel-28
Quality-score regression	<code>audit_quality.py</code>	—
4-class speaker decoder	<code>aad_4way.py</code>	CCA mel-28 + LightGBM
LOSO transfer	<code>aad_loso.py</code>	CCA mel-28

17.3 Success criteria (NeurIPS-ready)

- Pooled within-subject binary AAD ≥ 0.70 with CI clear of chance, *after* regressing out gaze to rule out the eye-tracking-in-disguise hypothesis.
- Clean per-subject tier behaviour; no more than one subject (S1) dropped and that exclusion justified by a hardware-quality score rather than post-hoc accuracy.
- A single clearly-novel contribution beyond the dataset itself — most likely the synchronised gaze-EEG fusion plus the per-trial comprehension questionnaires, both of which are absent from existing AAD datasets.